

**EFFECT OF HERBICIDE ON ROOT ROT OF
CHICKPEA CAUSED BY *Rhizoctonia bataticola***

THESIS

**Submitted to
Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola
In partial fulfilment of the requirements
for the Degree of**

**MASTER OF SCIENCE
IN
AGRICULTURE
(PLANT PATHOLOGY)**

**By
BORKAR SIMA HIVRAJ**

Dr. PDKV Library, Akola

581.2/BOR



157056

**DEPARTMENT OF PLANT PATHOLOGY
POST GRADUATE INSTITUTE, AKOLA**

**DR. PANJABRAO DESHMUKH KRISHI VIDYAPEETH,
KRISHINAGAR PO, AKOLA (MS) 444104**

Enrolment Number – GG/14

2014

DECLARATION OF STUDENT

I hereby declare that the experimental work and its interpretation of the Thesis entitled "**EFFECT OF HERBICIDE ON ROOT ROT OF CHICKPEA CAUSED BY *Rhizoctonia bataticola***" or part thereof has neither been submitted for any other degree or diploma of any University, nor the data have been derived from any thesis / publication of any University or scientific organization. The source of materials used and all assistance received during the course of investigation have been duly acknowledged.

Place: Akola


(BORKAR SIMA HIVRAJ)

Date: 3/4/2014

Enrolment No. GG-14

CERTIFICATE

This is to certify that, the thesis entitled "EFFECT OF HERBICIDE ON ROOT ROT OF CHICKPEA CAUSED BY *Rhizoctonia bataticola*" submitted in partial fulfillment of the requirement for the degree of "Master of Science in Agricultural (Plant Pathology)" of Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola is a record of bonafide research work carried out by BORKAR SIMA HIVRAJ under my guidance and supervision.

The subject of the thesis has been approved by the Student's Advisory Committee.

Place: Akola

Date: 9/7/2014



(V.R. GUPTA)

Chairman
Advisory committee

Countersigned



Associate Dean
Post Graduate Institute, Akola
Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola

THESIS APPROVED BY THE STUDENT'S ADVISORY COMMITTEE
INCLUDING EXTERNAL EXAMINER (AFTER VIVA-VOCE)

1. Chairman Dr. V. R. Gupta

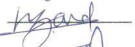
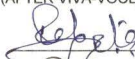
2. Member Dr. S. S. Mane

3. Member Dr. G .K. Giri

4. Member Dr. V.V. Goud

5. External Member

(B. R. W) Gupta



ACKNOWLEDGEMENT

After a long period of sincere hard work, the time has arrived when. I take up the opportunity to thanks all those, whose who directly took part in shaping this unique piece of work.

I feel immense pleasure in taking this opportunity of expressing my profound, humble indebtedness and deepest sense of gratitude and thanks from core of my heart towards my respected guide Dr. V. R. Gupta, Plant Pathologist, Pulses Research Unit, Dr. PDKV, Akola for his valuable guidance, constant inspiration, keep interest, kind advice and constructive criticism, right from selection of research problem up to final shaping of the thesis in the present form.

It is my proud privilege to express my deep sense of gratitude towards members of my advisory committee, Dr. S. S. Mane, Head, Department of Plant Pathology, Dr. G.K. Giri, Associate Professor, Department of Plant Pathology, Dr. V. V. Goud, Assistant Professor of Agronomy, Dr. PDKV, Akola for their intellectual stimulation, Kind suggestions and comments during investigation.

I wish to record my cordial thanks to Dr. S. S. Mane, Head, Department of Plant Pathology, Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola. I am grateful to him for providing all necessary facilities during the course of investigation.

My gratitude goes to Dr. D. M. Mankar, Associate Dean, PGI, Dr. PDKV, Akola for providing all necessary facilities during my course of study. I also take this opportunity of expressing my sincere thanks to, Dr. B. T. Raut, Professor and Ex Head, Dr. R. M. Gade, Professor, Department of Plant Pathology, Dr. R. W. Ingle, Associate Professor, Department of Plant Pathology, and Dr. S. T. Ingle, Assistant professor, Dept. of Plant Pathology, Dr. PDKV, Akola.

It is my proud privilege to record my deepest sense of gratitude and cordial thanks to affectionate Mrs. Manjusha Gaikwad SRA, Shri. S. H. Wavre, Shri. R. S. Ghawade, Shri. M. S. Joshi JRA, Shri. K. K. Kakade, Agril.

Assistant and all staff members of Department of plant pathology PGI, Dr.PDKV, Akola for their valuable suggestions and comments during my research investigation.

I also express my sincere thanks to all my senior Y. K. Belkar, Dr. P. N. Rakhonde, Amol Shitole, and Atul Gawande, who helped me a lot during to the course of investigation.

I have no words to express my regards and love to my friends who play a vital role in our life. I would like to express my feeling of indebtedness to classmates and my friends. I gratefully acknowledge the most willing help and co-operation from all my friends for their timely help and moral support at various points during the course of this study.

No words or phrases can convey my exact feeling to my parents for their pathetic efforts, sacrifices, encouragement in educating me at the cost of their comfort and consolation. I humbly express deepest sense to my Grandfather respected, Mr. Wasudeoji Langewar and my Grandmother Sau. Vandana Langewar, Father Shri. Hivraj borkar, Mother Sau. Alka Borkar, Uncle Shri. Anil borkar, Anty Sau. Sindhu borkar, Sisters Priynka and Monu, Brothers Nitin and kishor, My relatives Mr. Amit Pimparkhede, who took pains in providing all my needs, invoking divine blessing, guidance and shadowing me by showing their back to sun without which this work would not have been seen the light of the day at all.

I am also thankful to Mr. Prakash Mohite (M/s. Shree Grafix, Akola) for their neat setting for preparing this dissertation.

I am very much thankful to all authors and researchers, whose articles helped me in organizing my research work on proper line and utilize proper tools for interpretation of the results.

I was fortunate enough to receive the kind co-operation from almost every one way or the other, during my stay at this institute. It is extremely difficult to thank all of them individually by name; this short coming may please be pardoned.

Last but never least, it is difficult to list all those to whom I express my gratitude to my beloved God "shri. Ganapati Bappa", "shri. Gajanan Maharaja" and "Maa Durga" whose grace is always to motivate and inspire me for every movement of my work.

Place: Akola


(Borkar Sima Hivraj)

Date: 9/07/2014 **Enrolment No.** GG-14

Table of Contents

Sr. No.	Particulars	Page
A	List of Tables	i
B	List of Figures	ii
C	List of Plates	iii
D	List of Abbreviations	iv
F	Thesis Abstract	v
I	Introduction	1-5
II	Review of Literature	6-22
III	Material and Methods	23-33
IV	Results and Discussion	34-52
V	Summary and Conclusions	53-55
VI	Literature cited	56-62
	Vita	
	Appendix	



(A)

List of Tables

Table	Title	Page
1.	Pathogenicity test of <i>Rhizoctonia bataticola</i> on variety of Chaffa-816	35
2.	<i>In vitro</i> effect of fungicides and herbicides against <i>R.bataticola</i> at 25 ⁰ C Temperature and different pH ranges. (Poisoned food technique)	36
3.	<i>In vitro</i> effect of fungicides and herbicides against <i>R.bataticola</i> at 30 ⁰ C Temperature and different pH ranges. (poisoned food technique)	38
4.	<i>In vitro</i> effect of fungicide and herbicide against <i>R.bataticola</i> at 35 ⁰ C Temperature and different PH ranges (poisoned food technique)	40
5.	<i>In vitro</i> effect of fungicide and herbicide against <i>R.bataticola</i> at 40 ⁰ C Temperature and different PH ranges. (poisoned food technique)	42
6.	<i>In vitro</i> Compatibility of <i>Trichoderma viride</i> with herbicides. (poison Food technique)	44
7.	<i>In vitro</i> antagonism of <i>Trichoderma viride</i> against <i>R.bataticola</i> (dual culture technique) with herbicides.	44
8.	Population of <i>Rhizoctonia bataticola</i> in different treatments.	46
9.	Efficacy of different treatment in maintaining the Population of <i>T.viride</i>	48
10.	Efficiency of different treatment against root rot of chickpea (7DAS) observed at DAS	50
11.	Percent Root rot at 30 DAS and 60 DAS day after sowing in Pot culture	52

(B)

List of figures

Figures	Title	After Page
1.	View of experimental layout	30
2.	<i>In vitro</i> effect of fungicides and herbicides against <i>R. bataticola</i> at 25°C Temperature and different pH ranges. (Poisoned food technique)	37
3.	<i>In vitro</i> effect of fungicides and herbicides against <i>R. bataticola</i> at 30°C Temperature and different pH ranges. (poisoned food technique)	38
4.	<i>In vitro</i> effect of fungicide and herbicide against <i>R. bataticola</i> at 35°C Temperature and different PH ranges (poisoned food technique)	40
5.	<i>In vitro</i> effect of fungicide and herbicide against <i>R. bataticola</i> at 40°C Temperature and different PH ranges (poisoned food technique)	42
6.	Population of <i>Rhizoctonia bataticola</i> in different treatments	46
7.	Efficacy of <i>Trichoderma</i> in different treatment	48
8.	Effect of different treatment on root rot of chickpea at different interval	50
9.	Percent Root rot at 30 DAS and 60 DAS day after sowing in Pot culture	52

(C)

List of plates

Plate	Title	After Page
1.	General view of experimental Field	30
2.	Pure culture and Pathogenicity test of <i>Rhizoctonia bataticola</i>	35
3.	<i>In vitro</i> Compatibility of <i>Trichoderma viride</i> with herbicides (Poison Food technique)	43
4.	<i>In vitro</i> antagonism of <i>Trichoderma viride</i> against <i>R.bataticola</i> (dual culture technique) with herbicides.	44

(D) List of Abbreviations

%	:	Per cent
/	:	Per
µg	:	Micro-gram
µl	:	Micro-liter
°C	:	Degree centigrade
Conc.	:	Concentration
DAI	:	Days after inoculation
DAS	:	day after sowing
et al.	:	<i>Et al.</i> ia (and associates)
Fig.	:	Figure
g	:	Gram
hrs	:	Hours
i.e.	:	id est (that is)
ml	:	Mili liter
<i>in vitro</i>	:	in laboratory
<i>in vivo</i>	:	in field
PDA	:	Potato dextrose agar
SE (m) ±	:	Standard error
Sig.	:	Significant
NS	:	Non-significant
Spp.	:	Species
ST	:	Seed treatment
Viz.,	:	Vide licet (namely)

(F)

THESIS ABSTRACT

- a) Title of thesis : "EFFECT OF HERBICIDE ON ROOT ROT OF CHICKPEA CAUSED BY *Rhizoctonia bataticola*"
- b) Full name of student : Borkar Sima Hivraj
- c) Name and address of Major Advisor : Dr. V. R. GUPTA
Assistant Professor,
Pulses Research Unit
Dept. of Plant Pathology
Dr. P.D.K.V, Akola
- d) Degree to be awarded : M.Sc. (Agriculture)
- e) Year of award of degree : 2014
- f) Major subject : Plant Pathology
- g) Total number of pages : 62
- h) Number of words in the Abstract : 351
- i) Signature of the student : 
- j) Signature, name and address of forwarding authority :



Head
Department of Plant Pathology
PGI, Dr. PDKV, Akola

ABSTRACT

The research work entitled "Effect of herbicide on root rot of chickpea caused by *Rhizoctonia bataticola*" was carried out in field and pots culture during 2013-2014 at Pulse Research Unit and Department of Plant Pathology, Dr .PDKV, Akola.

The fungicides viz. Carbendazim @ 0.1% and Thiram@ 0.2% gave the cent per cent growth inhibition of *Rhizoctonia bataticola* and the herbicide Pendimethalin @100µl/100ml media was found very effective against *Rhizoctonia bataticola* and recorded inhibition of *R. bataticola* at 40⁰ C

in between 60.99 to 68.15 % at different pH ranges i.e. 5.5,6.5,7.5 and 8.5. But at the same time it showed 77.64% per cent growth inhibition of *Trichoderma viride* also. In dual culture technique performed on poison food Pendimethalin inhibited *Trichoderma viride* by 61.11%. where as that of *Rhizoctonia bataticola* by 46.05%.

Seed treatment with Thiram@ 2 g/kg +Carbendazim@ 1 g/kg of seed coupled with soil application of *Trichoderma* @ 2.5 kg/ha recorded minimum population of *Rhizoctonia bataticola* in soil ($0.33, 0.66$ and 1.00×10^4 cfu/g soil) at 15 , 30 and 45 days of sowing thereby inhibiting 84.75 per cent population over control.

The average population of *Trichoderma viride* was highest (16.5×10^4 cfu/g soil) in the treatment of *Trichoderma* through soil @ 2.5 kg /ha and was lowest in soil application of Pendimethalin @ 1.0 kg a.i./ha (PE) i.e. 1.33×10^4 cfu/g soil followed by seed treatment with *Trichoderma viride* @ 4g /kg seed + Soil application of Pendimethalin @ 1.0 kg a.i./ha (PE) i.e. 1.77×10^4 cfu/g soil.

Maximum root rot reduction (77.59%) was achieved due to combine application of Thiram @ 2 g/kg + carbendazim @ 1 g/kg as seed treatment and *Trichoderma viride* as soil application. Pendimethalin alone could reduce the root rot by 13.31%.

In pot experiment, 100 per cent reduction of root rot was observed in seed treatment with Thiram@ 2 g /kg +Carbendazim @ 1 g /kg followed by seed treatment with @ 2 g /kg +Carbendazim @ 1 g /kg + soil application of *Trichoderma* @ 2.5 kg/ha i.e. 95.00%. The root rot reduction was by 75.60% due to seed treatment of *Trichoderma viride* @ 4 g/kg.

CHAPTER I

INTRODUCTION

1.1 Background information

Chickpea (*Cicer arietinum* L.) is one of the most important legumes grown in Asia. It is commonly known as *channa*, gram, *garbanzo bean* etc. It belongs to family *Leguminaceae* and sub family *Papilionaceae*. It is a self pollinated, diploid ($2n=2 \times 16$) with genome size $1C = 740\text{Mbp}$. In Asia, India is the largest producer of Chickpea, contributing more than 70 percent of the total world and ranks first in the Indian Subcontinent. The centre of Origin is in Eastern Mediterranean.

In India Chickpea is cultivated mainly as rain fed in *Rabi* season. The Madhya Pradesh, Rajasthan, Uttar Pradesh, Maharashtra, Karnataka, Andhra Pradesh, West Bengal and Haryana are the major chickpea growing states which contribute about 93 per cent toward the total acreage.

It is consumed in many forms, as Dal, Chhole. The flour is also used in preparations of various types of sweets and snacks. It's nutritionally rich, conferring 17-23 per cent protein, 46 per cent carbohydrates, 4-5 per cent fats, 6 per cent crude fiber, 3 per cent ash and also rich in calcium, iron and vitamin C. Being a legume crop it also improves the soil fertility by fixing the atmospheric nitrogen. Chickpea meets 80% of its nitrogen (N) requirement from symbiotic nitrogen fixation and can fix up to 140 kg N ha^{-1} from air.

Globally, India has been the pioneer in chickpea supply, leads in Chickpea Production with 8.25 million tones under area of 9.21 million ha and average yield of 896 kg/ha . In Maharashtra, it is grown on 1.29 million ha with production of 1.11 million tones and yield of 863 kg/ha during the year 20011-12(DES-2013). Generally Chickpea is

cultivated as a rain fed crop in Maharashtra state and yield losses amounted to 10 to 15 percent.

Dry root rot caused by necrotropic fungus *Rhizoctonia bataticola* (Taub).Butler (Pycnidial stage: *Macrophomina phaseolina* (Tassi) Goid) is emerging as a serious threat to the chickpea production worldwide (Pande and Sharma, 2013).The disease is reported to be more severe when the crop is exposed to moisture stress condition. Previous studies have revealed and reported high levels of pathogenic and genetic variation in *R.bataticola* (RB) from different parts of the world and in India.

1.2 Importance of study

Chickpea is a source of human food and animal feed; it also helps in the management of soil fertility, particularly in dry lands and conserves natural resources which are essential for sustainable agriculture. Chickpea is a cheap source of protein, and supply 358 calories energy from 100g of seed, constituting about 99 percent in grain on dry weight basis and hence referred as “poor Man's Meats”.

It fixes atmospheric nitrogen in soils and thus improved soil fertility and saves fertilizer costs in subsequent crops. The crop can be grown as a second crop using residual moisture.It is an excellent source of Protein, fiber, complex Carbohydrates, Vitamins, and Minerals. The growing demand in both the domestic and export markets provides a source of cash for small holder producers. It increases livestock productivity as the residue is rich in digestible crude protein content compared to cereals.

Some of the serious disease in order of their importance is wilt *Fusarium oxysporum f. sp. ciceri*), wet root rot (*Rhizoctonia solani*), dry root rot (*Rhizoctonia bataticola*) Ascochyta blight (*Ascochyta rabiei*) and collar rot (*Sclerotium rolfsi*). However, the crop is attacked by a large number of fungal, bacterial and viral diseases of

which root rot caused by *Rhizoctonia bataticola* f. sp. *ciceri* is major limiting factor in production.

The disease generally appears around flowering and podding stage. The whole plant dries up and turns straw-colored. Roots become black and brittle and dry.

Although high temperature and low soil moisture is known to favors the disease, there has been a great deal of increase in the use of herbicides in the world during the past 30 years. Little attention is given to the possibility that they may also be toxic to plant-associated microorganisms such as those that cause disease stress and those that protect crops against diseases. Herbicides are reported to affect the incidence and severity of plant diseases (Altman and Campbell, 1977).

The herbicides have impact on soil-residing plant pathogenic microorganisms that cause disease in plants and on root-associated beneficial microorganisms that suppress the activity of plant pathogens (biocontrol agents). Those herbicide-mediated changes in the activity of plant pathogens and biocontrol agents may result in an increase or a decrease in Pathogen-induced stress in plants.

Herbicides which are now a day used widely for weed management is not into the light to have any effect on *Rhizoctonia bataticola*. Its effect on *Rhizoctonia bataticola* is not well studied. Hence the present investigation was under taken with a view to study the following objectives.

1.3 Objectives of the study

- 1) To study the effect of pre- emergence herbicide on root rot of Chickpea *in vivo*.
- 2) To know the effect of pre- emergence herbicide on *Rhizoctonia bataticola* at different pH and temperature *in vitro*.

1.4 Scope and limitation

The *Rhizoctonia bataticola* causing root rot in chickpea is soil borne and may survive in soil in the form of Sclerotia for long time. The fungus has a wide host range and can infect the root of several crops this may reduces the yield of crops.

Several reviews in the last 30 years have cited cases where herbicides have either increased or decreased plant diseases (Katan and Eshel, 1973). New and more effective herbicides are continuously being developed to replace the old ones. In the course of their development, herbicides are routinely screened for the absence of phytotoxicity to crops.

We have been interested in the impact of herbicides on the activity of biocontrol microorganisms and on the intensity of pathogen-induced stress because the herbicide concentration varies drastically in the soil, microorganisms and plants are sensitive to changes in the levels of herbicides, and unlike other soil factors (e.g., temperature, moisture), the impact of herbicides on the activity of bio control microorganisms has virtually been neglected.

Herbicides are also known to cause changes in the incidence and severity of some types of pathogen-induced stress possibly by affecting plant pathogens. (Altman and Campbell, 1977).

1.5 Hypothesis

Being soil borne in nature the root rot disease is very difficult to control. Although some method of disease control through fungicide is available, the method of application of recommended fungicide i.e. carbendazim as soil drenching after appearance of disease is not practiced by farmers being not feasible on large area. However herbicides are now a days used by farmers without hesitation which is also applied in the soil before emergence of the seed. Keeping this in view, it was thought that the applied herbicide may have effect on root rot pathogen and compatibility with recommended fungicides that will be useful in the application of fungicide through soil also, apart from added advantage of herbicides effect on the pathogen.

CHAPTER II

REVIEW OF LITERATURE

The investigation on "Effect of herbicides on root rot of chickpea caused by *Rhizoctonia bataticola*" was carried out at Department of Plant Pathology (PGI) Dr. PDKV, Akola, (MS). pertaining to the effect of different seed dressing fungicides, herbicides and bio-control agents on root rot.

Attempts therefore made to putforth the pertinent recently published work review critically under the following major heads.

- 2.1 Pathogen
- 2.2 Symptomlogy
- 2.3 Fungicides
- 2.4 Herbicides
- 2.6 Role of temperature and pH
- 2.7 Bio-agents
- 2.8 Compatibility
- 2.9 Growth Parameter

2.1 Pathogen

Uppal et al. (1936) reported that two isolates of *R. bataticola* did not grow at 9°C and 11°C respectively after 2 days, but slight growth of both isolates was observed after 5 days. Similarly at 40°C, 3.9 cm and 1.9 gm mycelia mat was observed after 7 days, respectively. They also recorded the optimum temperature between 30 and 35°C for the isolates of *R. bataticola*.

Rangaswami et al. (1958) isolated the pathogen *Rhizoctonia bataticola* by tissue segment method on potato dextrose agar (PDA) medium.

Saksena and Vaartaja (1961) proved the pathogenicity of *Rhizoctonia* species under aseptic condition in test tubes and in partially sterile soil on tomato, lettuce, cabbage, beet, cedar, and elm and pine seedlings, showing no specialization.

Fiki et al. (1987) isolated the *Macrophomina phaseolina* more frequently (65.2%) followed by *Fusarium* spp. (25.0%), *Rhizoctonia solani* (7.6%) and *Sclerotium rolfsii* (2.2%) from different samples showing charcoal-rot and root rot symptoms.

Hooda et al. (1988) found that isolates of *R. bataticola* from different host or same host has shown wide variation in virulence and pathogenicity

Amusa et al. (2007) proved the pathogenicity of seed borne pathogen *Macrophomina phaseolina* (*Rhizoctonia bataticola*) by seed inoculation method and stated that pathogen shows the symptoms within 21 days in cowpea and soybean.

Aghakhani et al. (2009) found twenty - three isolates of *R. bataticola* causing dry root rot of chickpea (*Cicer arietinum*) from the samples collected from 10 different major chickpea growing states of India. The isolated were highly variable in their morphological and cultural characters as pathogenicity /virulence.

Andrabi (2011) could isolate *Fusarium oxysporum* f.sp.ciceri, *F.solani* and *Rhizoctonia solani* from the wilted chickpea (*Cicer arietinum*) plants.

Sundravadana et al. (2012) observed that isolated *R. bataticola* from black gram differed in their virulence. The shoot isolate showed maximum seedling mortality on black gram (16.10 %) and was significantly superior by green gram (15.29 %).The minimum seedling mortality was recorded on red gram (7.88 %) soybean (5.3 %) and cowpea (9.33 %).

2.2 Symptomology

Meyer et al. (1973) observed that fungus *M. phaseolina* is soil-borne and may survive as sclerotia.

Srivastav (1976) observed *Macrophomina phaseolina* pathogenic on cotton causing leaf spot and root rot.

Dhingra and Sinclair (1978) found that the dry root rot caused by sclerotial form of *Rhizoctonia bataticola* (Pycnidial form of *Macrophomina phaseolina*) causes upto 60 % yield loss. It is a diverse omnipresent soil borne species with a wider host range involving more than 500 species in 75 families.

Chopra and Sharma (1986) reported the pre and post emergence mortality caused by *Rhizoctonia bataticola*.

Haware (1990) observed that drying of the plants appear suddenly in the field at flowering and podding. There was no drooping which is commonly seen in the plants affected by Root rot. The leaves and stems of affected plants are straw-colored. The roots are usually dry, unless the soil is wet. The tap root is quite brittle, show shredding of the bark and can be broken easily. Minute dark black sclerotial bodies can be seen with the aid of a 10X hand lens on the surface of the root, as well as in the pith.

Sharma and Pande (2013) reported that the Dry root rot (DRR) caused by *Rhizoctonia bataticola* (Taub.) Butler. [Pycnidial stage: *Macrophomina phaseolina* (Tassi) Goid] was a potentially emerging constraint to chickpea production than wilt (*Fusarium oxysporum* f. sp. *ciceri*). Increasing incidence of DRR indicated strong influence of climate change variables such as temperature and moisture on the development of disease.

2.3 Fungicides

Menten et al. (1976) studied the effect of three fungicides on the mycelium growth of *M. phaseolina* and concluded that Benomyl promoted a greater mycelial growth inhibition than Pentachloronitrobenzene (PCNB) and Carboxin.

Goyal and Mehrotra (1981) tested the 9 fungicides against *R. bataticola* *in vitro* on chickpea. Benlate, Bavistin, Thiophanate methyl and Dyrene were promising. In inhibiting the fungal growth.

Chauhan (1986) found that the Carbendazim and quintozene were superior against *Rhizoctonia solani* and *Rhizoctonia bataticola* causing seedling disease of cotton with maximum germination when seeds were treated with Carbendazim

Taya et al. (1990) conducted green house tests in different ways and found that the best control of *R. bataticola* was given by Carbendazim alone or in combination with Thiram as seed treatment, presowing, soil- drench and seed treatment + drenching after sowing the treated seed of chickpea.

Peshney et al.(1992) reported that the Thiram, Capton, PCNB, Moncozeb Iprodione Carboxine, Carbendazim, Thiobendazol (all @ 0.2%)and Tridemorph @ 0.07% were effective in controlling the mycelial growth and at the concentration of 0.006% effective in controlling sclerotial germination of *Rhizoctonia bataticola in vitro*. Carbendazim and Thiobendazol were most effective and inhibited growth at the Concentration of 0.006%..There was no reduction in sensitivity towards Carbendazim or Thiobendazol.

Jeyalakshmi et al. (1999) concluded that the *Trichoderma viride*, *Pseudomonas fluorescens* and Carbendazim were effective for controlling seed borne pathogens viz., *Fusarium* spp. and *Rhizoctonia* spp.

Lambhate et al. (2002) observed *in vitro* efficacy of different Fungicides against *Macrophomina phaseolina* (*Rhizoctonia bataticola*) and found that Bavistin, Ridomil, Topsin M @ 0.1, 0.2 and 0.3% were effective to inhibit the growth of *M. phaseolina*.

Sindhan et al. (2002) concluded that the Carbendazim was most effective causing 94.2 and 100% inhibition in mycelial growth and sclerotial production respectively of *Rhizoctonia bataticola*.

Mathur (2006) tested the different systemic and non-systemic fungicides against *Macrophomina phaseolina* (*Rhizoctonia bataticola*) rot in cowpea and reported that the Thiophanate-methyl 75 %WP, Carbendazim 50% WP, Tebuconazole 2 DS, Tebuconazole 250 WE, Hexaconazole 5% EC and Thiram 75 DS completely inhibited fungal growth at 50 ppm, followed by Chlorothalonil 75 WP which proved next in efficacy.

Vijay-Mohan et al. (2006) conducted fungicidal trials on management of dry root rot of chickpea caused by *R. bataticola*. Carbendazim (0.2%) and Etaconazole (0.1%) as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.6 and 18.2 per cent, highest grain yield of 192 and 18.9 q/ha respectively.

Rai et al. (2007) reported that 5 to 10 ppm concentration of Carbendazim was lethal to *Trichoderma viride* and inhibited mycelia growth.

Suryawanshi et al. (2008) tested seven fungicides against *Macrophomina phaseolina* in mungbean and found that Carbendazim was most effective both in field and lab condition.

Prabhu (2009) found that Carbendazim seed treatment showed much reduction in population of *M.phaseolina* followed by Thiram seed treatment during both the years. In Carbendazim seed treated plots, the initial population of *M.phaseolina* was 36cfu/g dry weight of soil, and it get reduced to 34.20 cfu on 30 DAS and

continued to decrease gradually and attained 25.20cfu/g of soil on 90DAS.

John Priya et al. (2010) reported that the Thiophanate-methyl and Carbendazim were most effective against the mycelial growth of *Rhizoctonia bataticola* followed by Carboxin and Captan.

Kumar and Khare (1990) found Bavistin and Vitavax most effective, showing 100% inhibition of mycelial growth of *Rhizoctonia bataticola* at 50 ppm followed by kitazin which showed 100% inhibition at 100 ppm. Thiram, Blitox and Captan were the least effective.

Nagamma et al. (2012) evaluated different fungicides against *Macrophomina phaseolina* (*Rhizoctonia bataticola*), among which Thiram @ 0.3%, Carboxin + Thiram @ 0.2% and Carbendazim @ 0.1% showed 100% percent inhibition of the pathogen which causes dry root rot in chickpea.

Rosane (2013) evaluated the active ingredients Carbendazim and Penflufen + Trifloxystrobin were the most powerful ones to control *M. phaseolina*, and were the most efficient in soybean seed treatments.

2.4 Herbicides

Smith et al. (1964) reported that the biological activity of herbicides extends beyond their effect on target organisms and herbicides may influence plant-pathogen interactions through their effect on the pathogen, the plant, or on the surrounding soil organisms including symbiotic interactions.

Katan and Eshel (1973) studied the four mechanisms which cause an increase in disease as the Herbicide may directly influence pathogen growth, virulence of the pathogen, host susceptibility, and/or change the relationships between other pathogens and other organisms.

Mohammad (1973) studied the effect of 10, 100 and 1000 ppm of seven different nitro- herbicides, namely DNOC, Dinoseb, Nitrofen, Fluorodifen, Trifluralin, Nitralin and Cobex on the growth of *Rhizoctonia solani*, *Fusarium oxysporum*, *Sclerotium bataticola*, and *Aspergillus niger* and observed that Trifluralin and Cobex gave only moderate action at the lowest concentration, and a significant inhibition when used in higher concentrations (100-1000 ppm). DNOC and dinoseb gave an inhibitory effect only at high concentrations.

Altman and Campbell (1977) found that the herbicides affect the incidence and severity of plant diseases.

Canaday et al. (1986) observed the increased soybean root colonization by *Macrophomina phaseolina* (Tassi) Goid due to herbicide-induced stress. Herbicides Pendimethalin and Trifluralin inhibited mycelial growth of *Pythium arrhenomones* and root colonization of *Macrophomia phaseolina*.

Bowman and Sinclair (1989) noted decreased soybean seedling vigour in *R. Solani* infested soil treated with the herbicides Alachlor, Choramben, Dinoseb, Fluchloralin, or Naptalam compared to seedlings growing in soil not treated with a herbicide in the greenhouse.

Singh and Kulshrestha (1991) studied the microbial degradation of Pendimethalin in vitro and observed that *Fusarium oxysporum* and *Paecilomyces varioti* two soil fungi, in culture media degraded Pendimethalin into two metabolite. *Rhizoctonia bataticola* decomposed Pendimethalin yielding only 3, 4- Dimethyl -2, 6- dinitroaniline metabolite. Fungal decomposition of Pendimethalin involved nitro reduction and dealkylation.

Bauske and Kirby (1992) reported that the soybean growing in soil treated with several different dinitroaniline herbicides did not have increased severity of root and hypocotyl rot caused by *R.*

solani in field and greenhouse conditions, compared with untreated soil.

Russin et al. (1995) observed that the Alachlor and Metolachlor have reduce the fungal colony diameter and increase the microsclerotia production of *M. phaseolina*, a fungus causing charcoal rot in sorghum. Although alachlor and metolachlor reduced germination of microsclerotia but did not affect fungal colonization of sorghum roots

Black et al. (1996) found that the paraquat acts in the presence of light and the herbicidal activity increases with increased light intensity. Inhibitory effects of Paraquat on mycelial growth of pathogens were reported in *Rhizoctonia solani*.

Mushtaq (1997) evaluated ten herbicides for their effects on seed- and soil-borne fungi in sandy loam and medium black soils. In *in vitro* tests, Methabenzthiazuron (50 micro g/ml) supported maximum germination (90.7%) of *Rhizoctonia bataticola*. In field experiments sown with soybeans and linseed, 20 days after herbicide application the saprophytic mycoflora in both soil types predominated the pathogenic population although the efficacy of the herbicides varied with the soil type. Benthicarb, Fluchloralin and Oxyfluorfen showed a significant increase in fungal population in sandy loam soil but not in medium black soil.

Dissanayake et al. (1998) observed that the Pendimethalin and Trifluralin inhibited mycelial growth of *Pythium arrhenomones* and root colonization of *Macrophomia phaseolina*.

Heydari and Misaghi (1998) found that in pot tests, damping-off caused by *R. solani* was better controlled by fungicides Captan and Mounsirin, when the soil was treated with paraquat or simazine. Both herbicides increased the toxicity of fungicides against mycelial growth of the pathogen may be due to an increased concentration of ROS in the plant.

Zidan et al. (1998) observed that Superphosphate suppressed Tolclofos-methyl or Thiram fungi toxicity while Pencycuron/Dichlofluanid mixtures showed greater fungitoxicity when mixed with herbicides.

Bradley et al. (2002) observed that herbicides did not significantly decrease the *Rhizoctonia solani* disease severity index compared to control, but did cause some increase in disease severity compared to control. In greenhouse studies, Dimethenamid + Metribuzin, Pendimethalin, Acifluorfen, and Imazethapyr caused an increase *Rhizoctonia* rot and hypocotyls rot severity compared to the no-herbicide control.

Harikrishnan and Yang (2002) found Positive effect of Glyphosate on vegetative growth of several *Rhizoctonia solani* isolates and anastomosis groups. However, the herbicide influenced the production of fruiting bodies of this pathogen. The number of sclerotia produced was higher but these sclerotia remained smaller in the presence of the herbicide compared to the untreated control. Further they observed that root rot severity differed among treatments in two cultivars of soybean, In BSR 101 *R. solani* + Pendimethalin + Imazethapyr treatment significantly reduced plant stand compared with the *R. solani* + Pendimethalin treatment. In P 9344, only treatment *R. solani* + Pendimethalin revealed a significant increase in root rot severity compared with the *R.solani* treatment.

Chen Hou De et al. (2005) reported that Fenoxaprop-p-ethyl, Butachlor, Fluroxypre, Chlorsulfuron +Isoproturon, Tribenuron-methyl, Dicamba and Metsulfuron-methyl inhibited mycelial growth in *R.Ceralis* at varying degrees, and the EC₅₀ values for these herbicides were 7.03,26.38,67.20,88.05,199.35,1249.55 and 203.31 mg liter⁻¹,respectively.The rates of the Inhibition of Pathogenicity of the pathogen were 44.83, 25.00, 1.00%, 31.03,35.34 and 25.86% respectively, where the herbicides were applied at 50mg liter⁻¹.Microscopic observation indicated that the mycelia treated with

herbicides were deformed at varying degrees. Fenoxaprop - p delayed the invasion by the mycelia and inhibited the mycelia further penetrating the vicinal cells of Wheat.

Jha and Sharma (2006) screened the antifungal properties of Alachlor, Fluchloralin, Metolachlor, Pendimethalin, Isoproturon, Metribuzin and Oxyfluorfen *in vitro* against isolates of *Rhizoctonia bataticola* by incorporating the herbicides in molten Asthana and Hawker's medium at 100 and 200 ppm concentrations and found that Oxyfluorfen and Isoproturon inhibited the mycelial growth and affected sclerotial morphology more than the other herbicides tested.

Rai et al. (2007) studied the effect of Pendimethalin, Anilofos, Paraquat, Butachlor, Isoproturon, Alachlor and 2, 4-D at 25, 50, 100 or 500 ppm each on the growth of *R. solani*. Most of the herbicides retarded the growth of the pathogen by poisoned food technique

Saxena et al. (2014) screened seven herbicides, 2, 4-D ethyl ester, Pretilachlor, Anilofos, Alachlor, Butachlor, Fluchloralin and Pendimethalin and found that the herbicides were compatible with the test antagonist even at higher concentration (250 μ l a.i. /ml). They opined that the present results would be helpful to delineate the possibility of combining *T. harzianum* PBT 23 as biocontrol agent with agrochemicals under integrated pest management approaches.

2.5 Role of Temperature and pH

Jackson (1965) reported that pathogen could grow over a wide temperature range of 10°C to 45°C but the optimum temperature for its growth was found to be 30°C. The next best temperature for its growth was recorded as 35°C. Statistically the growth of the pathogen gradually decreased both below and above the optimum temperature (30°C).

Katariya et al. (2007) reported that the antifungal activity of Fluchloralin and Alachlor against *P. butleri* and *R. solani* causing damping off of cowpea is temperature-dependent. Sporangia production was completely inhibited at 50 μ l a.i./l and 100 μ l a.i./l of Pendimethalin and Fluchloralin, respectively. Regarding fungistatic action of certain herbicides against soil-borne fungi, Fluchloralin and Alachlor considerably reduced the growth of *Pythium butleri* and *R. solani*.

Patel and Patel (1990) reported 35°C to be optimum temperature for growth and sclerotial formation of *M. phaseolina*.

Singh and Kulshrestha (1991) observed that the effects of temperature were evident at both the concentrations (1 and 10 μ l a.i./l) of Fluchloralin and Pendimethalin. Both the herbicides proved most effective at 20°C and least inhibitory at higher temperature (30°C). There was a decrease in inhibition of mycelial growth on either side of this maximum inhibitory temperature (20°C) and it was more pronounced in Pendimethalin at 15°C as compared to 25 and 30°C.

Jha and Sharma (2006) observed variations in fungal biomass accumulation at different pH. Maximum biomass accumulation was observed between pH 6.0 to 7.0, which was in accordance with results obtained while studying effect of pH on the growth of *M. phaseolina* isolated from okra.

Csondes et al. (2007) reported 25-35°C as optimum temperature for the growth of *M. phaseolina* isolates collected from Hungary. The isolates of various climatic regions in Italy exhibited optimum temperature of 30-35 °C for their growth.

Vanita Salunkhe et al. (2009) found maximum mycelium growth and Sclerotial formation of *Rhizoctonia bataticola* at 35°C, minimum mycelial growth at 20°C and nil or rare sclerotial formation at 20°C and 40°C.

Sundravadana et al. (2012) observed the effect of temperature on mycelial growth of *R. bataticola* after 3 DAI and found

that 15°, 20° and 25°C temperature had recorded a mean mycelial growth of 26.83, 34.23 and 48.51 mm, respectively. The maximum mycelial growth i.e 74.76 and 74.20 mm was recorded at 30° and 35°C respectively.

Sharma and Pande (2013) observed that higher temperature range of 25 to 35 °C favored the growth of *M. phaseolina* isolated from pearl millet, sesame, horse gram and moth bean, 18.03% and 34.43% of isolates had max. RGR (62.04 and 53.71 mm/day), respectively at 25 °C while min. RGR (20.58 mm/day) was observed in only 4.92% of isolates. Similarly, at 30 °C, RGR varied from min. (38.57 mm/day) to max. (67.44 mm/day) with frequency distribution of 11.48% and 19.67%, respectively.

Singh and Nene (1990) isolated sixty-one isolates of *Macrophomina phaseolina* and studied for their adaptability to different temperature and pH. higher relative growth rate was observed at 30°C followed by 35°C. Similarly, higher mean dry mycelial weight was recorded at pH 6 and 7.0 which clearly indicated the preference of isolates to particular range of pH.

Reddy et al. (1990) reported that the three isolates of *M. phaseolina* differed in response to various concentrations across the three temperature regimes. Among the three isolates, TN 410 was the most sensitive for both GlyCry (GR50 = 7.74 mm) and Gly-K salt (GR50 = 0.25 mm) at 30 °C.

2.7 Bio-agents

Srivastava et al. (1976) observed that *T. harzianum* isolate 25 and *Pseudomonas fluorescens* isolate 4 significantly reduced the germination of sclerotia by 60% in natural field soil.

Kumar and Khare (1990) reported the antagonistic relationship of *T. harzianum* and *Bacillus subtilis* with *R. bataticola* and *Sclerotium rolfsi* in soybean. It was inferred that the population

increased and decreased due to the antagonistic activity of *T. harzianum* and *Bacillus subtilis*.

Haram et al. (1996) concluded that *Trichoderma harzianum* is an efficient biological agent that is commercially produced to prevent development of several soil seed pathogenic fungi like *Rhizoctonia bataticola*, *Fusarium Solani*, *F. oxysporum*.

Majumdar et al. (1996) observed antagonistic property of *Trichoderma* isolates against *M. phaseolina*.

Sindhan et al. (2002) reported that *Trichoderma viride* is the best antagonist inhibited 62.3 and 87.0% mycelium growth and sclerotia production followed by *Trichoderma harzianum*, *A. ochraceous*, *Bacillus subtilis* and *P.citrinum*, *A. niger* proved to be least effective in comparison to other antagonists but were superior to control.

Singh and Nene (1990) observed the efficacy of *Trichoderma harzianum*, *T. Viride*, *T. hematum*, *Pseudomonas fluoresces* and *Bacillus subtilis* in controlling *Rhizoctonia bataticola* causing dry root rot of chickpea *in vitro* and in field experiment. *T. harzianum* recorded the highest control of the pathogen both *in vitro* and *in vivo*.

Bandyopadhyay et al.(2003) found inhibited growth of *Rhizoctonia bataticola* by 51.1 per cent due to a strain of *Trichoderma* under *in vitro* conditions.

Chirame and Padule (2005) observed that the *Trichoderma viride* was superior for controlling the *Fusarium moniliformae*, *Macrophomina phaseolina* (*Rhizoctonia bataticola*) while *Trichoderma viride* was superior for controlling the *Fusarium moniliformae*, *Macrophomina phaseolina* (*Rhizoctonia bataticola*) and *Colletotrichum capsici* isolated from cotton seed.

Karthikeyan et al. (2006) demonstrated culture filtrates of *T. viride* inhibiting the growth of the pathogen as well as sclerotial germination to greater extent.

Paul et al.(2008) reported that among 11 *Trichoderma* isolates tested, *Trichoderma harzianum* showed higher inhibition of the growth of *Rhizoctonia solani* by 77 per cent under *in vitro* conditions.

Kaushal (2008) found that *T. harzianum* was effective in inhibiting the mycelial growth of *R. bataticola* the causal organism of chickpea dry root rot.

Choudhary et al.(2010) tested four bioagents viz., *Trichoderma viride*, *Trichoderma harzianum*, *Aspergillus versicolor* and *Bacillus* for the control of dry root rot in mungbean. In dual cultures, *Trichoderma viride*, *Trichoderma harzianum* and *Aspergillus versicolor* were effective in inhibiting the growth of *Macrophomina phaseolina* to the extent of 61 to 65%.

Doley and Paramjit kaur (2012) followed the dual culture technique and found that *T. viride* had significant antifungal activities towards both the pathogens inhibiting the mycelial radial growth of *S. rolfsii* by 75% and *M. phaseolina* by 71.42%.

2.8 Compatibility

Reddy et al. (1990) reported 48% inhibition of *T. viride* due to the herbicide Pendimethalin (0.1%) by poison food technique.

Jhonson and Subramanyan (2002) found that Pendimethalin inhibited 48 per cent growth of *Trichoderma* spp. in laboratory.

Desai and Kulkarni (2004) reported that Alachlor, Carbenadazim, Chlorpyrifos, Glyphosate, Organomercurial, Thiram and Trifluralin were highly inhibitory to *T. harzianum* whereas comparative safer agrochemicals to bioagents were Captan,

Acephate, Atrazine and 2, 4-D sodium salt. Inhibitory effects increased with increase in concentrations from 500 to 2000 ppm.

Tiwari and Singh (2004) observed that the mycelial growth of *T. harzianum* was completely inhibited by Carbendizam and Hexaconazole @ 1500 ppm, whereas Copper oxychloride and Mancozeb @ 1500 ppm has shown 90 and 41 per cent inhibition respectively.

Naseema Beevi et al.(2005) studied the *in vitro* compatibility of *T. harzianum* with Mancozeb, Carbendazim and Copper Oxychloride and found that Carbendazim at 0.1 per cent completely inhibited the mycelial growth while Mancozeb and Copper Oxychloride showed compatibility with the antagonist at 0.2 and 0.1 per cent respectively.

Chattannavar (2006) tested *in vitro* the effect of five herbicides (Alachlor, Glyphosate, Klass, Pendimethalin and Paraquat dichloride) on *T. harzianum* The herbicides were applied at 50, 100, 200, 250, 500, 750 and 1000 ppm to 5 mm culture disc of the fungi. Alachlor showed highest inhibition at the lowest concentration, mean fungal growth inhibition was due to pendimethalin (63.73%).

Parab et al.(2009) studied the sensitivity of *T. harzianum* to different fungicides and the results indicated that all systemic fungicides were found completely inhibiting growth of *T. harzianum* whereas Zineb, Copper hydroxide, Mancozeb and Copper Oxychloride were safe for the growth of *T. harzianum*.

Sharma and Pande (2013) found that Fluchloralin, Oxadiazon and Pendimethalin adversely affect the growth of *Trichoderma spp.* by 42.22, 37.77 and 55.55 per cent even at 125 µl/ml, 250 µl/ml and 250 µl/ml respectively.

Amrutha et al. (2014) isolated ten *Trichoderma spp.* from chickpea rhizosphere and root entophytic region and tested against *Rhizoctonia bataticola*, Out of the ten isolates, isolate-7 showed

highest inhibition percentage (83.33). In compatibility tests with commonly used fungicides, *Trichoderma* isolate-7 showed highest compatibility with validamycin (72.22%) followed by Copper Oxychloride (66.66%).

2.9 Growth Parameter

Reddy et al. (1990) found that combination of herbicides Fluchloralin and Alachlor with certain fungicides like Carbendazim, Benomyl and Carboxin altered the fungicidal action of fungicides against *Fusarium oxysporum*, *Sclerotium rolfsii*, *Alternaria brassicicola* and *Colletotrichum capsici*. They expressed the view that the mixtures are at times synergistic and more effective.

Singh and Nene (1990) observed the higher yields in seed treatment with Carbendazim + Thiram followed by 2 sprays of Monocrotophas or Endosulfan.

Awadalla and Refai (1994) reported that Paraquat and Simazine inhibited the mycelial growth of *R. solani* and enhanced the inhibitory effects of the fungicides captan and mounsrin under *in vitro* conditions. The activity of certain fungicides (Tolclofos-methy/Thiram and Pencycuron), herbicides (Fluometuron and Pendimethalin) and their binary mixtures against soil borne fungi (*R. solani* and *C. rolfsii*) were evaluated *in vitro* and showed the physical compatibility between fungicides and herbicides. Mixture of fungicides and herbicides gave an antagonistic effect.

Sindhan et al. (2002) observed the Carbendazim 0.2% was most effective with 79.2% disease control and also significantly increased root and shoot length and dry weight in comparison to control.

Nagmani et al. (2011) conducted field experiment during 2009-2010 and 2010-11 *rabi* seasons at Andhra Pradesh, India to

devise an effective management strategy for dry root rot of chickpea. The seed treatment with Carbendazim at 2g/kg of seeds + seed treatment with *T. viride* at 4g/kg of seed + soil application of FYM fortified with *T. viride* (T₇) were the best treatment to decrease the mortality of root rot caused by *R.bataticola*. It also recorded highest growth parameters like shoot length and root length and also the highest yield.

Kamdi et al. (2012) found significantly increased yield per plot over control due to seed treatments. Carbendazim gave more seed yield (13.47q/ha) followed by *T.viride*+ Carbendazim (13.19q/ha), *T.viride*+Thiram (12.78q/ha) and *T.viride* alone (12.50q/ha) as against control (7.22q/ha).

CHAPTER III

MATERIAL AND METHODS

The present studies on "Effect of herbicide on root rot of chickpea caused by *Rhizoctonia bataticola*", and their aspects were conducted during 2013 – 2014 in the laboratory of Department of Plant Pathology, and field of Pulses Research unit, Dr. PDKV, Akola. The details of the material and methods used for investigation are given below.

3.1 Material

The following material was used during the present investigation.

3.1.1 Diseased samples

The Chickpea diseased plants were collected from experimental plot of Pulses Research Unit Dr. PDKV, Akola. The samples were examined for the presence of pathogen under research microscope and preserved for further investigation.

3.1.2 Glassware

The glassware used during the course of research work were of Borosil make. The glasswares viz., Petri plates, test tubes, conical flasks, measuring cylinder, glass rods, slides, cover slips, beaker, and pipettes were used during experimentation.

3.1.3 Culture media

Common laboratory media i.e. potato dextrose agar was used for the isolation of the pathogen associated with root rot of chickpea.

3.1.4 Chickpea Seeds

Seeds of root rot susceptible variety Chaffa of chickpea were obtained from the Pulses Research Unit, Dr. PDKV, Akola.

3.1.5 Chemical used

The fungicides viz., Thiram, Carbendazim, herbicide like Pendimethalin and other chemicals such as streptomycin sulphate and mercuric chloride etc. were used during the course of research work which were obtained from Department of Plant Pathology and Pulses Research unit.

3.1.6 Procurement of bio-agents

The bio-control agent i.e. *Trichoderma viride* used in the studies was obtained from Department of Plant Pathology Dr. PDKV, Akola.

3.1.7 Equipments used

The laboratory equipments viz, Autoclave, laminar air flow, incubator, Refrigerator, pH meter, Freezer, cork borer, hot air oven Research microscope, etc, were used during the investigation.

3.1. 8 Miscellaneous material

Inoculation needles, forceps, polyethylene bags, distilled water, rectified spirit, spirit lamp, mercuric chloride, earthen pots, Labels etc. were used during the course of the present investigation.

3.1.9 Precaution to eliminate contamination

All the isolation and inoculation work were carried out in laminar air flow under aseptic condition. The table top of laminar air flow was sterilized by glowing ultraviolet light for half an hour prior to commencement of work. The working surface of laminar flow and side glasses were surface sterilized with denaturation spirit. Moreover, other

such necessary care was taken to maintain and carryout work under aseptic condition.

3.1.10. Soil samples

Soil samples were collected from rhizosphere of Chickpea experimental plot of Pulses Research Unit Dr. PDKV, Akola. Samples were air dried and kept in polythene bags and further used for counting colonies of pathogen.

3.2 Methods

3.2.1 Preparation of culture media

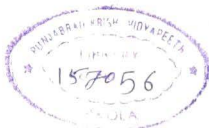
Potato dextrose agar (PDA) medium was prepared as per their composition used for isolation and maintenance of cultures of *Rhizoctonia bataticola*.

3.2.2 Disinfection / sterilization of laboratory materials

Glass Petri plates, pipettes, beakers and test tubes were washed with cleaning powder under running tap water and sterilized in hot air oven at 180°C for 1 hour before use. Distilled water and PDA medium were sterilized in an autoclave at 1.04 kg /cm² for 15 min. The sterilized media was used for experimentation.

3.2.3 Isolation of *Rhizoctonia bataticola*

The infected root and stems were washed with running tap water to remove any soil remains and then cut into small pieces before being dipped in sodium hypochlorite solution (2%) for two minutes for surface sterilization. These plant parts were then passed through changes of distilled water, dried between sterilized filter paper, then placed on medium with and without antibiotics. The plates were incubated at 25°C±1. The growth obtained on bits was transferred on PDA.



3.2.4. Mass Multiplication of culture

The sorghum grains were used for the mass multiplication of *Rhizoctonia bataticola* in laboratory. The sorghum grains were soaked in water for overnight. These grains were spread on the clear blotter paper to dry. The grains were taken in the tray and Calcium carbonate @ 25 g/kg was added to grains to avoid the clumping of grains in flask. About 300gm moistened grains filled in each 1000 ml flask. On plugging of conical flask, plugs were wrapped with paper and autoclaved for 30 min. at 15 lbs psi pressure for two times by keeping 24 hours gap between two autoclaving (Sterilization) i.e. carried out at two successive days. The grains were then inoculated with the pure culture of *Rhizoctonia bataticola* in aseptic condition and incubated at $27\pm 2^{\circ}\text{C}$ for 5 days. The flask's were shaken to avoid the clumping of grains and to facilitates early growth of the fungus on the grains .The grains turn blackish due to mycelial growth of the *Rhizoctonia bataticola*.

3.2.5 Preparation of sick soil and Pathogenicity test

Soil was put in gunny bags and sterilized in autoclave at 1.05 kg/cm^2 for 1 hour consequently for 3 days. The mass multiplied pathogen was added in 1:10 proportion to soil and thoroughly mixed. Thus soil was made sick.

The plastic pots of size 15cm diameter were taken and surface sterilized with 0.1%Mercuric chloride. The sick soil was filled in sterilized pots $1/4^{\text{th}}$ of its capacity. The pots were watered lightly and incubated for 4 days .Chickpea seeds of chaff were sown (20 seeds per plot).The pots were monitored for seedling mortality. The root rot symptoms were recorded with an interval of 10 days after sowing. Similarly, four pots were kept without inoculation which served as control .Re-isolation of fungus was done from roots of root rot seedlings on PDA media.

3.3. Poisoned food technique.

3.3.1 *In vitro* effect of Herbicides and fungicides on *Rhizoctonia bataticola* at different temperature and pH.

To evaluate the effect of herbicides and fungicides on *Rhizoctonia bataticola*, the poison food technique was used. Potato dextrose agar medium was prepared, equally distributed measuring 100ml in 250ml conical flask and pH was adjusted as, 8.5, 7.5, 6.5 and 5.5 and the media was sterilized in autoclave. Requisite quantity of each of the fungicide and herbicide and their combination was added in sterilized melted (45°C) PDA separately, so as to obtain desired concentration. Flask containing poisoned medium was shaken well to have even and uniform distribution of chemical. About 20ml of melted poisoned PDA was poured in each sterilized petriplate and allowed to solidify. These petriplates were inoculated by test fungus separately. Six mm disc of one week old fungus culture was cut with a sterilized cork borer, lifted and transferred aseptically in the centre of a petriplate containing the medium poisoned with test fungicide and herbicide. The control plates were kept where the culture disc were grown in same condition on PDA without fungicides and herbicides. Treated plates were incubated at different temperature ranges like 25, 30, 35 and 40°C for a period of seven days. Colony diameter was recorded in mm and per cent of mycelial inhibition was calculated as per formula given below based on the average of colony diameter. The data of mycelial growth was also subjected to statistically analysis and conclusions were drawn.

$$PI = \frac{C - T}{C} \times 100$$

Where,

PI = Per cent inhibition

C = Growth in control plate in mm

T = Growth in treatment plates in mm

3.3.2 *In vitro* Compatibility of *Trichoderma viride* with herbicides

Pure culture of *Trichoderma viride* was taken. Compatibility tests were conducted under *in vitro* condition to find out compatibility of herbicides with *Trichoderma viride*. For this herbicide Pendimethalin was evaluated by poison food technique. Herbicide was added to molten PDA just before pouring. The PDA plates amended with herbicides were inoculated aseptically with *Trichoderma* by transferring five mm diameter agar disc from fresh culture. Three replications were maintained for each treatment. Unamended PDA serve as the control. Inoculated petri plates were incubated at $25\pm1^{\circ}\text{C}$. The radial growth of *T. viride* was measured after seven days and compared with control. The percent inhibition of *T. viride* was calculated based on the diameter of growth of the colony by using the formula.

$$I = \frac{C - T}{C} \times 100$$

Where,

I = Per cent inhibition

C = Growth of fungus in control (mm)

T = Growth of fungus in treatment (mm)

3.4 Combined efficacy of *T.viride* and herbicide against *Rhizoctonia bataticola*

3.4.1 Dual culture and Poisoned food technique

Trichoderma viride was tested for their biocontrol properties against *Rhizoctonia bataticola* alongwith herbicide Pendimethalin by combination of dual culture and poison food technique. Lawn culture of test fungi and bioagent *Trichoderma viride* were prepared in petriplate. The mycelial disc of 6mm diameter cut from the margin of 5 days old culture of both test pathogen and antagonistic were placed opposite to each other on PDA incorporated with herbicide in petriplates (90mm). The distance between block was 6

cm. The petriplate with disc of *Rhizoctonia bataticola* served as the control. The inoculated petriplates were incubated at $27 \pm 2^{\circ}\text{C}$ in BOD incubator for 7 days. The growth of *Rhizoctonia bataticola* was measured and the percent growth inhibition of intersecting colonies was calculated as per following formula.

$$PI = \frac{C - T}{C} \times 100$$

Where,

PI = Per cent inhibition

C = Growth of *R. bataticola* in control plate in mm

T = Growth of *R. bataticola* in treatment plates in mm

3.5 Field Experiment

3.5.1 Experiment details

The study was conducted by using seeds of Chaffa-816 root rot susceptible variety of Chickpea under field condition. The experimental details are given below.

1. Year of experiment : Rabi 2013.
2. Design of experiment : Randomized Block Design(RBD)
3. No .of Replications : Three
4. Crop : Chickpea
5. Variety : Chaffa-816
6. Spacing : 30×10 cm
7. Plot size(Gross) : 2.4×4 m
8. Number of treatments : 10 (as below)

3.5.2 Treatment details

T ₁	control
T ₂	Seed treatment with Thiram @ 2g/kg+ carbendazim@1g/kg of seed.
T ₃	Seed treatment with <i>Trichoderma</i> @ 4g /kg.
T ₄	Application of Pendimethalin@1.0 kg a.i./ha(PE)
T ₅	Seed treatment with Thiram @ 2g/kg+ carbendazim@1g/kg+ Application of Pendimethalin.@1.0 kg a.i./ha.
T ₆	Seed treatment with <i>Trichoderma</i> @4g/kg+Application of Pendimethalin @1.0 kg a.i./ha(PE)
T ₇	Soil application of <i>Trichoderma</i> @ 2.5kg/ha.
T ₈	Soil application of <i>Trichoderma</i> @2.5kg/ha +Application of Pendimethalin@1.0 kg a.i./ha.
T ₉	Seed treatment with Thiram @ 2g/kg+ carbendazim@1g/kg+ Soil application of <i>Trichoderma</i> @ 2.5kg/ha.
T ₁₀	Seed treatment with Thiram @ 2g/kg+ carbendazim@1g/kg+ Soil application of <i>Trichoderma</i> @ 2.5kg/ha.+ Application of Pendimethalin@1.0 kg a.i./ha.

3.5.3 Plan of layout

The treatments were statistically randomized and the plan of layout of field and pot experiment was prepared (Fig 1 and Plate 1).

3.5.4 Land preparation

The soil of experimental plot was deeply ploughed and harrowed twice to pulverize the soil before start of experiment and the



Fig. 1. View of experimental layout

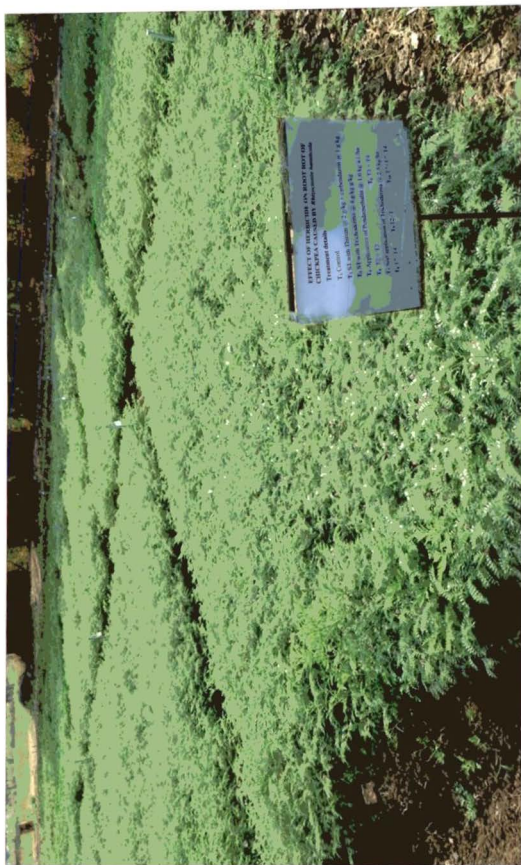


Plate 1. General view of experimental field

mass multiplied culture of *Rhizoctonia bataticola* on sorghum grains was incorporated in the line at the time of sowing.

3.5.5 Observation recorded

3.5.5.1 Measurement of disease incidence

Percent Root rot incidence were recorded at 15 days interval as DAS based on symptoms and the percent disease incidence was calculated by using the formula mentioned below.(Singh and Nene,1990)

$$\text{PDI} = \frac{\text{No. of dead seedlings}}{\text{No. of total seedlings}} \times 100$$

3.6 . Population count of *R.bataticola* from soil.

3.6.1 Preparation of Potato Dextrose Agar media (PDA)

The medium was prepared by using following ingredients.

Peeled Potato	200g
Dextrose	20g
Agar	20g
Distilled water	1000ml

3.6.2 Counting colonies of *R.bataticola* by serial dilution method

The initial dilution was prepared by adding 1 g of air dry soil sample from composite sample into 9 ml distilled sterilized water and dilution blank labeled 10^{-1} . Thus diluting original sample 10 times (1:10).Vigorously shaken the dilution on a magnetic shaker for 2-3 minutes to obtained uniform distribution and release of micro-organism from adhering soil particles. Allowed the soil particles to settle and from the first dilution transferred 1 ml of suspension to the dilution blank number of 10^{-2} by sterile pipette, shaken well for minutes. From

10^{-2} suspension transferred 1 ml of suspension to 10^{-3} dilution blank with a fresh sterile pipette. Repeated the procedure till the original sample had been diluted to 10^{-7} using every time a fresh sterile pipette. From the appropriate dilution (10^{-4} and 10^{-5}) transferred 1 ml of suspension with respective pipette to sterile petriplates. Three petriplates were used for each dilution. Approximately 20 ml melted medium added to each petridishes, containing the diluted sample. Mixed the content of each dish by rotating gently to distribute the cell through out the medium. Allowed the plates to solidify and incubate these plates in inverted position for 2-7 days at $28 \pm 2^{\circ}\text{C}$. The 4th and 5th dilution were plated in petriplates containing selective medium for isolating *R.bataaticola*. Well separated individual black colonies were marked and detected.

3.7 Pot culture experiment

3.7.1 Experiment details

A pot culture experiment was conducted in green house condition in Completely Randomized Design (CRD) with same treatments as laid in field with 3 replications on Chickpea Variety Chaffa-816. The Inoculums of test Pathogen *Rhizoctonia bataticola* were mass cultured on crushed Sorghum seeds and were added to the soil@100g per kg of soil. Prior to use, the earthen pots were disinfected with 0.1% Mercuric chloride solution, the seeds of Chaffa variety were treated with Carbendazim, Thiram, *Trichoderma viride* as per treatment details and Pendimethalin as soil application (PE). Then seeds were sown @ 20 seeds per pot and the pots were watered lightly. The pots were kept in cage house and the seedlings mortality was recorded at 25 days after sowing.

3.7. 2 Observation recorded

3.7.2.1 Germination of seeds

For estimation of percent germination total no of germinated seeds were selected from three replication and per cent germination was calculated as per treatment.

3.7.2.2 Measurement of disease incidence

In the pots, disease plants were observed at 30 DAS and 60 DAS. Number of diseased plants were recorded treatment wise and the total number of plants for calculating per cent root rot.

3. 8 Statistical analysis

The statistical analysis of the data was done by statistical method as suggested for 'F' test significance. The standard error (SE) and critical difference (CD) at 1% level of probability were calculated for lab. Experiments and at 5% for field experiment. Values of critical difference were used to interpret the result. The data have been illustrated graphically at appropriate place in the text.

RESULTS AND DISCUSSION

The research work entitled "Effect of herbicides on root rot of chickpea caused by *Rhizoctonia bataticola*" was carried out during *Rabi*, 2013-2014 in the field of Pulses Research Unit , and laboratory of Department of Plant Pathology, Dr.Panjabrao Deshmukh Krishi Vidyapeeth,Akola with objective to assess the effect of herbicide on root rot of chickpea caused by *Rhizoctonia bataticola* *in vitro* and *in vivo*.

4.1. Isolation of *Rhizoctonia bataticola*

Rhizoctonia bataticola was isolated from infected roots of chickpea showing typical root rot symptoms. Isolations were made at 10 days interval from 10th day of sowing upto 90th day. The association of *Fusarium oxysporum*, *Macrophomina phaseolina* and *Sclerotium rolfsii* was predominant. Andrabi (2011) isolated *Fusarium oxysporum*, *F.solani* and *Rhizoctonia solani* from the wilted chickpea (*Cicer arietinum*) plants. The higher frequency was observed with *R. bataticola* under the present studies.

4.2. Morphology of *Rhizoctonia bataticola*

Mycelium of *Rhizoctonia* were superficial or immersed, hyaline to brown, branched, septate, often tree like in form. The basic characters of the genus were formation of sclerotia. Sclerotia were dark brown to black,smooth,hard,and were formed of thick walled cells of irregular size but of uniform texture. The cells of the hyphae were barrel shaped, branching more or less at right angles and pale brown colour.

4.3. Pathogenicity test of *Rhizoctonia bataticola*

The pathogenicity test was performed by soil inoculation technique. The no. of wilted plants per pot in sick soil were recorded and presented in Table1 and Plate 2.

The data revealed that in sick soil *Rhizoctonia bataticola* caused 100% root rot on Chaffa-816 in 30 DAS. The typical root rot symptoms were observed within 30 DAS as compared to control where no root rot was observed. The pathogen from the infected plants was reisolated and compared with original culture and found identical, thus Koch's Postulates were proved. Singh and Nene (1990) also isolated the pure culture of *R. bataticola* (*Macrophomina phaseolina*) from infected chickpea, pigeonpea, pea, groundnut and sorghum plants from ICRISAT fields.

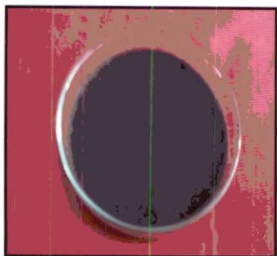
Table 1: Pathogenicity test of *Rhizoctonia bataticola* on variety Chaffa-816

Pathogen	<i>Rhizoctonia bataticola</i>	Control
Variety	Chaffa-816	Chaffa-816
No. of seeds sown	20	20
No. of seeds germinated	17	18
Percent seed germinated	85	90
Root rot infected plant	17	0
Percent root rot	100	0

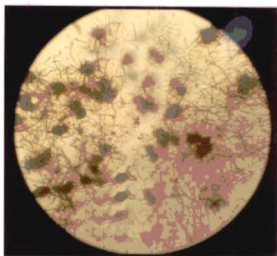
4.4. Efficacy of fungicides, herbicides, and bio agents against *Rhizoctonia bataticola* in vitro

4.4.1 Mycelial growth inhibition at 25°C temperature and different pH range

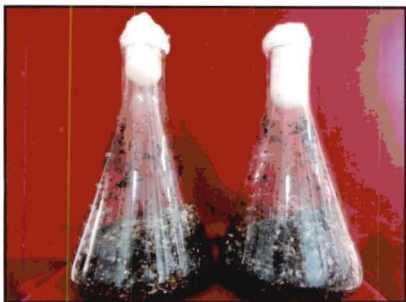
The activities of certain fungicides (Thiram and Carbendazim), herbicide (Pendimethalin) and their mixtures were evaluated against *Rhizoctonia bataticola* in vitro. Mixture of fungicides and herbicides gave synergistic effects against pathogen.



Petri plate with culture of *Rhizoctonia*



Sclerotia of *Rhizoctonia bataticola*



Mass culture of *Rhizoctonia bataticola*



Control



Showing yellow and drooping with infection of *Rhizoctonia bataticola*

The data in regards to effect of herbicide and fungicide on *R. bataticola* is presented in Table 2 and Fig. 2 which revealed cent per cent inhibition of *R. bataticola* due to combination of Thiram @ 0.2 per cent and Carbendazim @ 0.1 per cent in all pH at 7th DAI at 25°C. Similarly the combination of herbicide and fungicides i.e. Thiram+ Carbendazim +Pendimethalin showed cent per cent inhibition of pathogen at all four pH regimes i.e. 5.5, 6.5, 7.5 and 8.5 pH indicating the harmful ion concentration to restrict the growth at 25°C temperature. The pre-emergence herbicide i.e. Pendimethalin@ 100µl/100ml media alone was also effective in inhibiting the growth of pathogen compared to control and recorded 58.52, 55.23, 54.23 and 60.11 % inhibition at 5.5, 6.5, 7.5 and 8.5 pH regime. It indicated that in acidic to slightly alkaline pH though fungicides reduce the pathogen the herbicide Pendimethalin could also be able to minimize the root rot pathogen.

Inhibition of growth of *Rhizoctonia bataticola* and other fungi by herbicide pendimethalin is also reported by Jha and Sharma (2006), Rai et al. (2007), Russin et al. (1995), Katariya et al. (2007). Moreover few workers reported the most inhibitory effect of herbicide at 20°C and 6.0 to 7.0 pH viz., Singh and Nene (1990) and Jha and Sharma (2006) which supports the results of present findings.

4.4.2 Mycelial growth inhibition at 30°C temperature and different pH range

The data presented in Table 3 and Fig. 3 revealed 100 per cent inhibition of *R. bataticola* due to combination of Thiram @ 0.2 per cent and Carbendazim @ 0.1 per cent in all pH at 7th DAI. The combination of herbicide and fungicides Thiram+ Carbendazim +Pendimethalin showed cent per cent inhibition of pathogen at all four pH range i.e. 5.5 to 8.5 pH indicating the harmful effect of these chemicals to restrict the growth of pathogen at 30°C temperature.

Table 2. Effect of fungicides and herbicides against *R. bataticola* at 25°C temperature and different pH.

Treatment (for 100ml)	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition
Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP	5.5 pH		6.5pH		7.5pH		8.5pH	
	0.00	100	0.00	100	0.00	100	0.00	100
Thiram(0.2%) 50 WP+Carbendazim (0.1%)75WP +Pendimethalin 30%EC(100µl/100ml media)	0.00	100	0.00	100	0.00	100	0.00	100
Pendimethalin 30% EC(100µl/100 ml)	35.25	58.52	38.50	55.23	40.62	54.23	33.50	60.11
Control	85.00		86.00		88.75		84.00	
'F' test	Sig		Sig		Sig		Sig	
SE(M)±	0.1		0.2		0.1		0.1	
CD(P=0.01)	0.5		1.08		0.5		0.6	

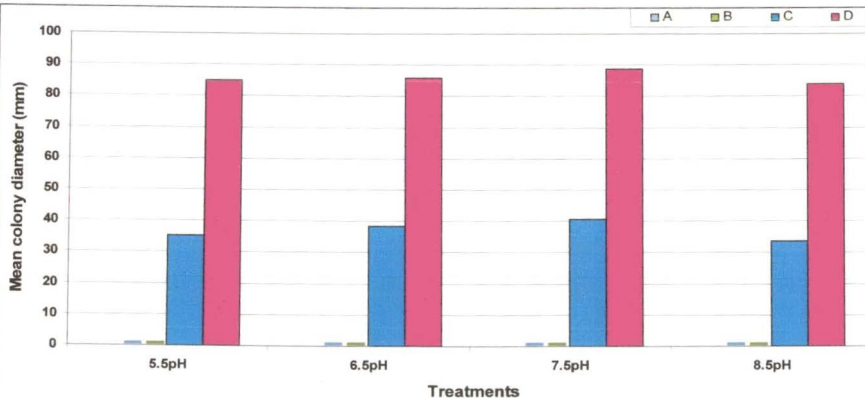


Fig. 1 : *In vitro* effect of fungicides and herbicides against *R. bataticola* at 25°C Temperature and different pH ranges. (Poisoned food technique)

A - Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP

B - Thiram(0.2%) 50 WP+Carbendazim 0.1%)75WP+Pendimethalin 30%EC(100µl)

C - Pendimethalin 30% EC(100µl)

D - Control

Table 3. Effect of fungicides and herbicides against *R.bataticola* at 30°C temperature and different pH.

Treatment	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition
Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP	5.5 pH		6.5 pH		7.5 pH		8.5pH	
	0.00	100	0.00	100	0.00	100	0.00	100
Thiram(0.2%) 50 WP+Carbendazim (0.1%)75WP +Pendimethalin 30%EC(100µl/100ml media)	0.00	100	0.00	100	0.00	100	0.00	100
Pendimethalin 30% EC(100µl/100 ml)	80.50	8.52	82.06	4.85	85.37	5.14	78.50	10.02
Control	88.00		86.25		90.00		87.25	
'F' test	Sig		Sig		Sig		Sig	
SE(M)±	0.1		0.2		0.1		0.1	
CD(P=0.01)	0.61		0.82		0.8		0.8	

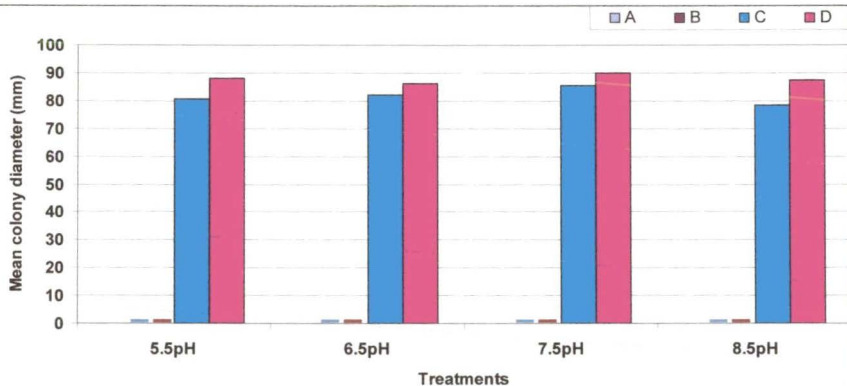


Fig. 2 : *In vitro* effect of fungicides and herbicides against *R.bataticola* at 30°C Temperature and different pH ranges (poisoned food technique)

A - Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP

B - Thiram(0.2%) 50 WP+Carbendazim 0.1%)75WP+Pendimethalin 30%EC(100µl)

C - Pendimethalin 30% EC(100µl)

D - Control

The pre-emergence herbicides i.e. Pendimethalin@ 100µl/100ml media was less effective in inhibiting the growth of pathogen compared to control and recorded 8.52, 4.85, 5.14, 10.02 % inhibition at 5.5, 6.5, 7.5 and 8.5 pH regime. Pendimethalin recorded inhibition between ranges of 4.85 to 10.02 % at different pH ranges. It indicated that in acidic to slightly alkaline pH at 30°C temperature herbicide Pendimethalin could be minimize the root rot pathogen compared to no treatment. But the inhibition was very less at this temperature.

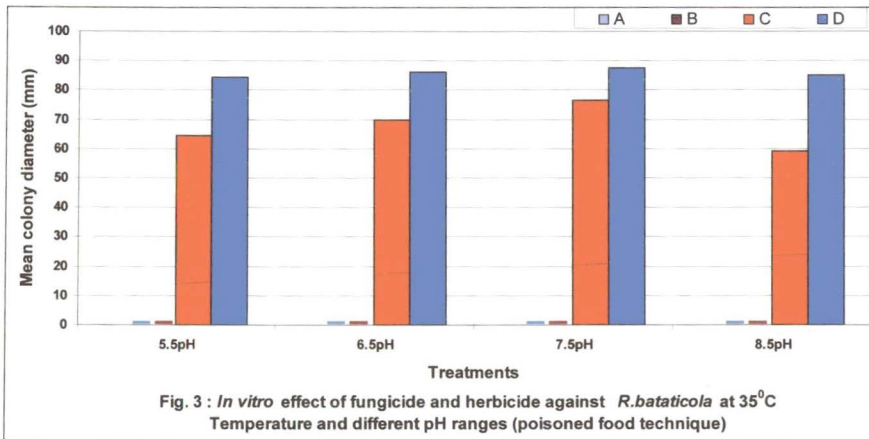
The less inhibitory effect of herbicide at this temperature might be due to the favourable temperature for the pathogen for its growth which can be correlated with the findings of Singh and Kulshetraha (1991) who also observed least effectivity of herbicides at higher temperature (30°C) More pronounced action was with Pendimethalin at 15°C as compared to 25 and 30°C when combined with fungicides.

4.4.3 Mycelial growth inhibition at 35°C temperature and different pH range

The 100 % inhibition of *R.bataticola* due to combination of Thiram @ 0.2 per cent and Carbendazim @ 0.1 per cent in all pH tests was observed which is presented in Table 4 and Fig. 4. The Pendimethalin@ 100µl/100ml media was least effective in inhibiting the growth of pathogen among all the treatments and recorded 23.45,18.60,12.57 and 30.29 % inhibition at 5.5, 6.5, 7.5 and 8.5 pH, respectively. This temperature also favours the growth of the pathogen as reported by Sundravada (2002) and Sharma and Pande (2013) and hence the inhibition at the temperature is less compared to 25°C.

Table 4. Effect of fungicide and herbicide against *R.bataticola* at 35°C Temperature and different pH.

Treatment	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition
Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP	5.5 pH		6.5 pH		7.5 pH		8.5 pH	
	0.00	100	0.00	100	0.00	100	0.00	100
Thiram(0.2%) 50 WP+Carbendazim (0.1%)75WP +Pendimethalin 30%EC(100µl/100ml media)	0.00	100	0.00	100	0.00	100	0.00	100
Pendimethalin 30% EC(100µl/100 ml)	64.50	23.45	70.00	18.60	76.50	12.57	59.25	30.29
Control	84.25		86.00		87.50		85.00	
'F' test	Sig		Sig		Sig		Sig	
SE(M)±	0.18		0.28		1.52		0.23	
CD(P=0.01)	0.82		1.24		0.35		1.03	



A - Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP

B - Thiram(0.2%) 50 WP+Carbendazim 0.1%)75WP+Pendimethalin 30%EC(100µl)

C – Pendimethalin 30% EC(100µl)

D – Control

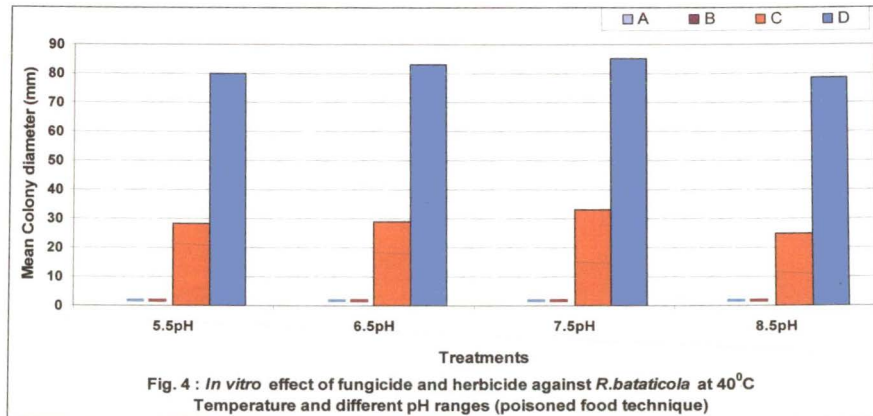
4.4.4 Mycelial growth inhibition at 40°C temperature and different pH range

Data indicated inhibition to the extent of 100% of *R.bataticola* due to combination of Thiram @ 0.2 per cent and Carbendazim @ 0.1 per cent at 40°C in all pH tested (Table 5 and Fig.5). The pre-emergence herbicide Pendimethalin@ 100µl/100ml media was also effective in inhibiting the growth of pathogen at this temperature. It showed 64.68, 65.06, 60.99 and 68.75 % inhibition at 5.5, 6.5, 7.5 and 8.5 pH, respectively. It indicated that at this temperature also in acidic to slightly alkaline pH along with fungicides, herbicide Pendimethalin could also minimize the root rot pathogen.

In the present study combination of two fungicides and fungicides + herbicide recorded 100 % inhibition of *R. bataticola* at all the temperature and pH range whereas herbicide alone could reduce the population of fungus upto 68 % (at 40° C and 8.5 pH) which indicated that herbicide can also reduce the *R. bataticola* to some extent and is not having any adverse effect on the performance of fungicides in combination, on the contrarary herbicide alone showed inhibition of pathogen which was encouraging and from this it can be taken into consideration that pendimethalin can increase the toxicity of fungicide as recorded by Heydari and Misaghi (1998). Reddy et al (1990) and Desai and Kulkarni (2004) also found that the herbicide and fungicides have synergistic effects sometimes. In the present study combination of fungicides alone had also given 100% inhibition of the pathogen and hence combining herbicide with these fungicides did not give the added advantage.

Table 5. Effect of fungicide and herbicide against *R.bataticola* at 40°C Temperature and different pH

Treatment	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition	Mean colony diameter (mm)	% growth inhibition
Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP	5.5 pH		6.5 pH		7.5 pH		8.5 pH	
	0.00	100	0.00	100	0.00	100	0.00	100
Thiram(0.2%) 50 WP+Carbendazim (0.1%)75WP +Pendimethalin 30%EC (100µl/100ml media)	0.00	100	0.00	100	0.00	100	0.00	100
Pendimethalin 30% EC(100µl/100 ml)	28.25	64.68	29.00	65.06	33.25	60.99	25.00	68.15
Control	80.00		83.00		85.25		78.50	
'F' test	Sig		Sig		Sig		Sig	
SE(M)±	0.31		0.35		0.26		0.25	
CD(P=0.01)	1.35		1.52		1.16		1.08	



A - Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP

B - Thiram(0.2%) 50 WP+Carbendazim 0.1%)75WP+Pendimethalin 30%EC(100µl)

C – Pendimethalin 30% EC(100µl)

D – Control

Mycelial growth inhibition of *R. bataticola* due to Thiram and Carbendazim are reported by many workers like Lambhate et al. (2002), Sindhan et al. (2002), Mathur (2006), Suryawanshi et al. (2008), John Priya et al. (2010), Peshney et al. (1992), Kumar and Khare (1990) and Nagma et al. (2012). As combination of fungicides alone gave 100% inhibition of pathogen, combining herbicide with fungicides did not give added advantage here however the inhibition of *R. bataticola* due to herbicide pendimethalin alone is supported by Awdalla and refai (1994), Mohammad (1973), Russin et al (1995) and Dissanayake and Canady (1998).

4.5. *In vitro* compatibility of pathogen, antagonist and herbicide

4.5.1 Effect of herbicide on *Trichoderma*

The potato dextrose agar media incorporated with Pendimethalin @100 μ l/100ml was used to grow *Trichoderma* for testing the compatibility of this chemical with bioagent. The results obtained are presented in Table 6 and Plate 3.

Data in Table 6 revealed that herbicide pendimethalin inhibited the growth of *Trichoderma* to the extent of 77.64 per cent where the colony diameter of *Trichoderma* in pendimethalin amended media was 20.15 mm compared to 90 mm in control at 7 day of incubation period. It indicated that *Trichoderma* is not compatible with the herbicide pendimethalin.

Reddy et al (1990), Chattannavar (2006) Sharma and Pandey (2013), Jhonson and Subramanyam (2002) also reported that Pendimethalin adversely affect the growth of *Trichoderma* spp. The result of present studies confirms their observations. However Saxena et al (2014) reported compatibility of pendimethalin with test antagonist. This contradictory result needs to know the name of the test antagonist to discuss with the present findings.

Table 6. Effect of herbicide on growth of *Trichoderma viride* (in vitro))

Treatment No.	Treatment details	Mycelial growth of <i>Trichoderma</i> (mm)	% inhibition of <i>Trichoderma</i>
T ₁	Pendimethalin @100µl/100ml media	20.15	77.64
T ₂	Control	90	0.00

DAI- Days after Inoculation

4.5.2 Combined effect of herbicide and *Trichoderma* on *Rhizoctonia bataticola*

In order to assess the ability of bioagent against pathogen in presence of herbicide this study was conducted by combining poison and dual culture technique. Pendimethalin was found to inhibit the *R.bataticola* to the extent of 46.05% in presence of *Trichoderma* and at the same time it inhibited the growth of *Trichoderma* to the tune of 61.11% (Table 7 and Plate 4). It showed that when both pathogen and antagonist are present herbicide had less inhibitory effect on antagonis as compared to alone presence of antagonist (Table 6). This results of adverse effect of herbicide on pathogen are in agreement with the findings Smith et al (1964), Katan and Eshel (1973), Chen HouDe et al. (2005) and Rai et al. (2007).

Table 7. Effect of Pendimethalin along with *Trichoderma viride* on *R.bataticola*

Treatment	Mycelial growth (mm)		% inhibition	
	<i>R.bataticola</i>	<i>T.viride</i>	<i>R.bataticola</i>	<i>T.viride</i>
Pendimethalin(100µl/100ml)	48.55	35.00	46.05	61.11
Control	90.00	90.00	0.00	0.00



Plate 3. Compatibility of *Trichoderma viride* with herbicides *in vitro* (poisoned Food technique)

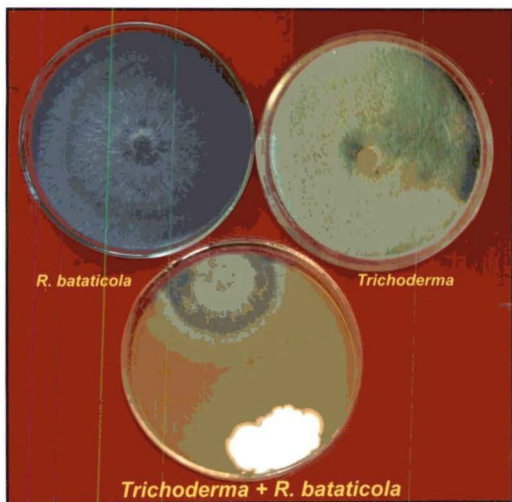


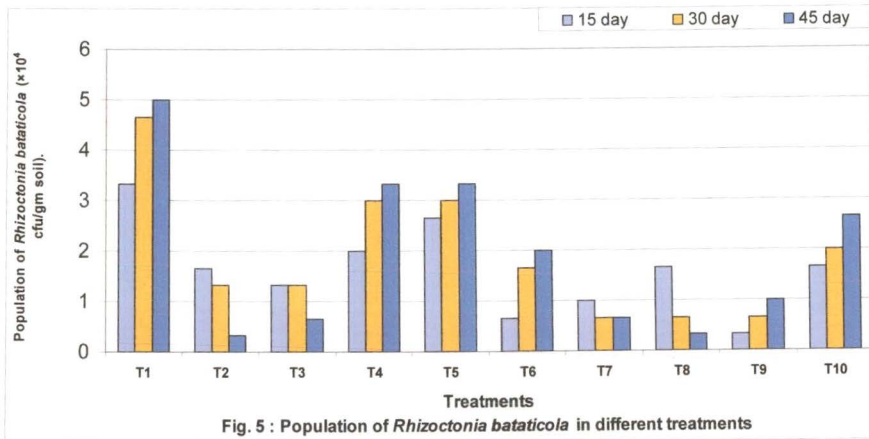
Plate 4. Antagonism of *Trichoderma viride* against *R. bataticola* (dual culture technique) with herbicides *in vitro*

4.6. Effect of different treatments on population of *R.bataticola* (in vitro).

The population of *R.bataticola* was recorded at 15 days of interval from the soil collected from experimental field and the data is presented in Table 8 and fig. 6. The highest average per cent reduction in population was observed in seed treatment of Thiram @ 2g/kg +Carbendazim @1g/kg + soil application of *Trichoderma*(@2.5kg/ha i.e. 84.75 % (0.66×10^4 cfu/g soil) followed by soil application of *Trichoderma* @ 2.5 kg /ha where the inhibition was 82.21 % (0.77×10^4 cfu/g soil). *Trichoderma*(@2.5kg/ ha(soil) +Pendimethalin@1.0 kg a.i./ha (PE) was the next best treatment showing inhibition of *R. bataticola* population by 79.67 % having average population of 0.88×10^4 cfu/g soil compared to 4.33×10^4 cfu/g soil in control. Fungicides applied to seed has fungitoxic effect, while the chemical herbicide might have activities to restrict the spore germination resulted in maintaining the lower population of pathogen. Thiram and Carbendazim results in continuous decline in population of *Rhizoctonia bataticola* from 30 to 90 DAS (Prabhu, 2009).

Table 8. Population of *Rhizoctonia bataticola* in different treatments.

Sr.No	Treatment	Population of <i>Rhizoctonia bataticola</i> ($\times 10^4$ cfu/gm soil).			Mean	% Reduction over control (Average)
		15 day	30 day	45 day		
1	Control	3.33	4.66	5.00	4.33	-
2	Thiram@2g/kg+Carbendazim@1g/kg(ST)	1.66	1.33	0.33	1.10	74.6
3	<i>Trichoderma</i> @ 4g /kg. (ST)	1.33	1.33	0.66	1.10	74.6
4	Pendimethalin@1.0 kg a.i./ha (PE)	2.00	3.00	3.33	2.77	36.03
5	Thiram@ 2g/kg +Carbendazim@1g/kg +Pendimethalin@1.0 kg a.i./ha (PE)	2.66	3.00	3.33	2.99	30.94
6	<i>Trichoderma</i> @ 4g /kg (ST)+Pendimethalin@1.0 kg a.i./ha (PE)	0.66	1.66	2.00	1.44	66.74
7	<i>Trichoderma</i> @2.5kg/ha (soil)	1.00	0.66	0.66	0.77	82.21
8	<i>Trichoderma</i> (@2.5kg ha(soil)+Pendimethalin@1.0 kg a.i./ha (PE)	1.66	0.66	0.33	0.88	79.67
9	Thiram@ 2g/kg +Carbendazim@1g/kg+ <i>Trichoderma</i> (@2.5kg/ha (Soil)	0.33	0.66	1.00	0.66	84.75
10	Thiram@2g/kg+Carbendazim@1g/kg+ <i>Trichoderma</i> (@2.5kg/ha +Pendimethalin@1.0 kg a.i./ha (PE)	1.66	2.00	2.66	2.10	51.50
	'F' test	Sig	Sig	Sig	Sig	
	SE(M) $\pm$	0.53	0.6	0.7		
	CD(P=0.01)	2.4	2.9	3.2		



A - Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP

B - Thiram(0.2%) 50 WP+Carbendazim 0.1%)75WP+Pendimethalin 30%EC(100 μ l)

C - Pendimethalin 30% EC(100 μ l)

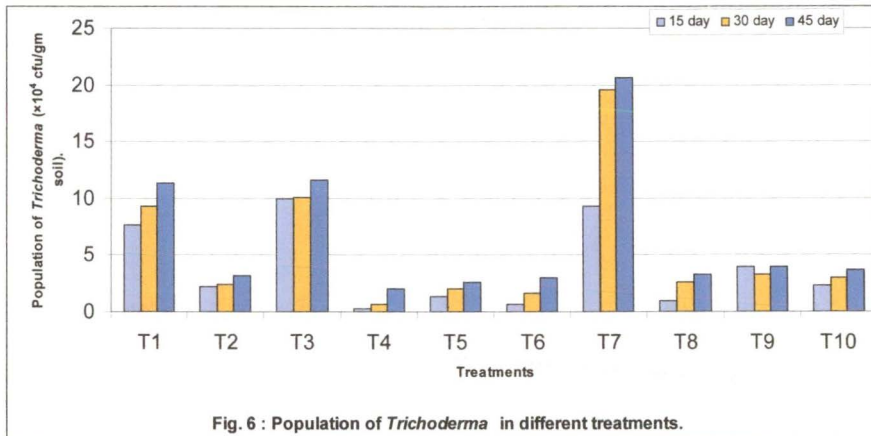
D - Control

4.7. Effect of different treatments on population of *T.viride* in vitro.

Highest population of *Trichoderma viride* (average, 16.5×10^4 cfu/g soil) was observed in treatment of soil application of *Trichoderma* @ 2.5 kg/ha when counted in laboratory from soil samples of experimental plot at different interval. (Table 9 and fig 7). The population of *Trichoderma* was lowest in treatment T₄-Pendimethalin application (PE) @ 1 kg a.i./ ha (1.33×10^4 cfu/g soil) followed by T₆ (1.77×10^4 cfu/g soil). In all the treatments gradual increase in population was observed. Chemicals either in the form of fungicide or herbicide showed deterrent effect on bioagent, thereby declining their population compared to control. Jhonson (2002) reported 48 per cent growth inhibition of *Trichoderma* spp. with Pendimethalin. These reports are in similar line with present observations. Fungicides applied to seed has fungitoxic effect, while the chemical herbicide might have activities to restrict the spore germination resulted in maintaining the lower population of antagonist.

Table 9. Population of *Trichoderma* in different treatments.

Sr.No	Treatment	Population of <i>Trichoderma</i> ($\times 10^4$ cfu/gm soil).			Mean
		15 day	30 day	45 day	
1	Control	7.66	9.33	11.3	9.43
2	Thiram@2g/kg+Carbendazim@1g/kg(ST)	2.25	2.4	3.15	2.6
3	<i>Trichoderma</i> @ 4g /kg. (ST)	10.00	10.11	11.66	8.66
4	Pendimethalin@1.0 kg a.i./ha (PE)	0.33	0.66	2.00	1.33
5	Thiram@ 2g/kg +Carbendazim@1g/kg +Pendimethalin@1.0 kg a.i./ha (PE)	1.33	2.00	2.66	1.99
6	<i>Trichoderma</i> @ 4g /kg (ST)+Pendimethalin@1.0 kg a.i./ha (PE)	0.66	1.66	3.00	1.77
7	<i>Trichoderma</i> @2.5kg/ha (soil)	9.33	19.6	20.6	16.5
8	<i>Trichoderma</i> @2.5kg ha(soil)+Pendimethalin@1.0 kg a.i./ha (PE)	1.00	2.66	3.33	2.33
9	Thiram@ 2g/kg +Carbendazim@1g/kg+ <i>Trichoderma</i> (@2.5kg/ha (Soil)	4.00	3.33	4.00	3.55
10	Thiram@2g/kg+Carbendazim@1g/kg+ <i>Trichoderma</i> (@2.5kg/ha +Pendimethalin@1.0 kg a.i./ha (PE)	2.33	3.00	3.66	2.99
	'F' test	Sig	sig	sig	Sig
	SE(M) $\pm$	1.4	2.2	2.5	
	CD(P=0.01)	6.7	9.0	10.6	



A - Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP

B - Thiram(0.2%) 50 WP+Carbendazim 0.1%)75WP+Pendimethalin 30%EC(100 μ l)

C - Pendimethalin 30% EC(100 μ l)

D - Control

4.8. *In vivo* effect of Herbicides, fungicides and bioagent on chickpea root rot caused by *R.bataticola*

4.8.1 Field experiment

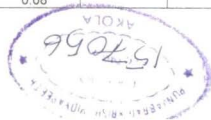
4.8.1.1 Effect of various treatments on per cent root rot in field

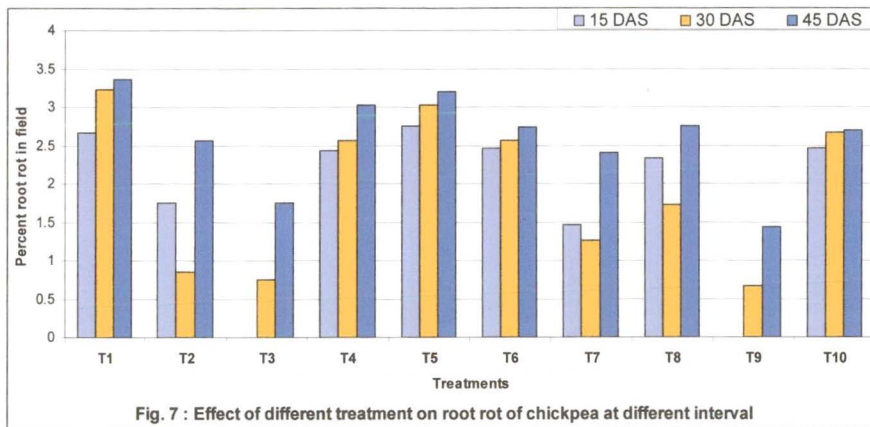
Treatment differences in regard to root rot incidence was significant and the data is presented in Table 10 and fig 8 .The observations were recorded at 15, 30 and 45 days after sowing. Maximum average root rot reduction was achieved due to combine application of Thiram @ 2 g/kg +Carbendazim @ 1 g/kg + *Trichoderma* @ 4 g/kg (77.59%) followed by soil application of *Trichoderma* @ 2.5 kg/ha (72.72%) and seed treatment with Thiram @ 2g/kg + Carbendazim @ 1 g/kg (43.83 %). Pre-emergence application of Pendimethalin alone @ 1 kg a.i./ha also showed the antifungal activities as root rot reduction was observed by 13.31 per cent. Seed treatment with *Trichoderma* alone recorded 36.31 per cent reduction in root rot. Efficiency of antagonist in the presence of herbicide was reduced, resulted in higher root rot incidence compared to application of *Trichoderma* alone either as seed treatment or through soil otherwise soil application of *Trichoderma* alone could reduce the disease by 72.72 per cent. Shrivastava et al (1976), Singh and Nene (1990) and Kaushal (2008) were of the opinion that *Trichoderma* can be used effectively for the management of root rot disease . The adverse effect of herbicide on *Trichoderma* was also observed under *in vitro* condition in present study as presented earlier. Although Heydari et al. (2007) recorded the higher antifungal activities of fungicide in presence of herbicides and the same results were observed under laboratory in present study but the results under field condition were different which might be due to adverse effect of herbicide on antagonist organisms also. Bardley et al. (2002) observed that herbicides did not significantly (Pp0.05) decrease the *Rhizoctonia* disease severity index (DSI) compared to the control, but did cause some increases in DSI compared to the control.

Table 10. Effect of different treatment on root rot of chickpea at different interval

Sr.No	Treatment	Percent root rot			Mean	% reduction over control
		15 DAS	30 DAS	45 DAS		
1	Control	2.66 (1.62)*	3.23 (1.79)	3.36 (1.83)	3.08	-
2	Thiram@2g/kg+Carbendazim@1g/kg(ST)	1.76 (1.32)	0.86 (0.76)	2.57 (1.60)	1.73	43.83
3	<i>Trichoderma</i> @ 4g /kg. (ST)	1.46 (1.20)	1.26 (1.12)	2.4 (1.54)	1.70	36.31
4	Pendimethalin@1.0 kg a.i./ha (PE)	2.43 (1.55)	2.56 (1.60)	3.03 (1.74)	2.67	13.31
5	Thiram@ 2g/kg +Carbendazim@1g/kg +Pendimethalin@1.0 kg a.i./ha (PE)	2.76 (1.65)	3.03 (1.74)	3.21 (1.79)	3.00	2.59
6	<i>Trichoderma</i> @ 4g /kg (ST)+Pendimethalin@1.0 kg a.i./ha (PE)	2.47 (1.57)	2.56 (1.60)	2.74 (1.65)	2.59	15.90
7	<i>Trichoderma</i> @2.5kg/ha (soil)	0.00 (0.00)	0.76 (1.76)	1.76 (1.32)	0.84	72.72
8	<i>Trichoderma</i> (@2.5kg ha(soil))+Pendimethalin@1.0 kg a.i./ha (PE)	2.33 (1.52)	1.73 (1.31)	2.76 (1.66)	2.27	26.29
9	Thiram@ 2g/kg +Carbendazim@1g/kg+ <i>Trichoderma</i> (@2.5kg/ha (Soil))	0.00 (0.00)	0.66 (0.53)	1.43 (1.19)	0.69	77.59
10	Thiram@2g/kg+Carbendazim@1g/kg+ <i>Trichoderma</i> (@2.5kg/ha +Pendimethalin@1.0 kg a.i./ha (PE)	2.46 (1.57)	2.66 (1.63)	2.7 (1.64)	2.60	15.58
	'F' test	Sig	Sig	Sig		
	SE(M)±	0.07	0.19	0.02		
	CD(P=0.05)	0.21	0.58	0.08		

(* Value are transformed by square root transformation)





A - Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP

B - Thiram(0.2%) 50 WP+Carbendazim 0.1%)75WP+Pendimethalin 30%EC(100µl)

C – Pendimethalin 30% EC(100µl)

D – Control

In this study maximum inhibition of root rot was found in seed treatment with fungicides like Thiram and Carbendazim. Inhibition of root rot due to Thiram and Carbendazim under field condition is also documented by Chauhan (1986), Taya et al. (1990), Vijay-Mohan et al. (2006) and Nagmani et al. (2011).

4.8.2 Pot experiment

4.8.2.1 Effect of various treatment on per cent Root rot in pot culture

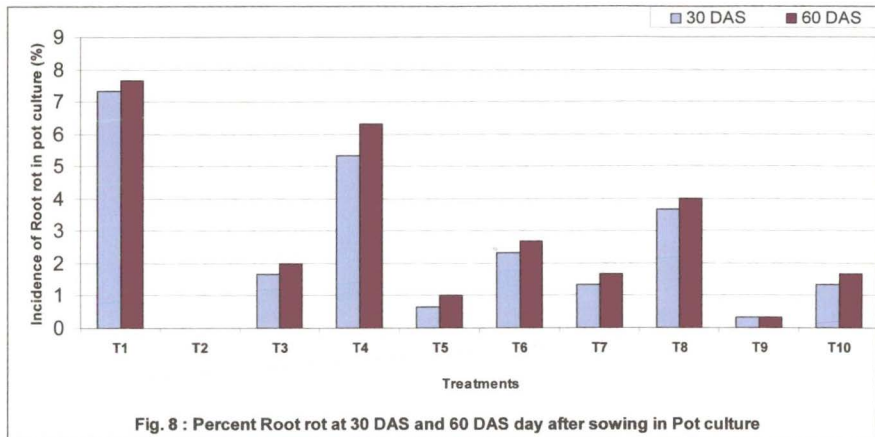
The data presented in Table 11 and fig. 9 showed 100 per cent reduction of root rot in seed treatment of Thiram @2 g/kg +Carbendazim @ 1 g/kg of seed followed by seed treatment with Thiram @2 g/kg +Carbendazim @ 1 g/kg +Soil application of *Trichoderma* @ 2.5 kg/ha (95.00%), Seed treatment with Thiram@ 2g/kg + Carbendazim @ 1 g/kg + Soil application of Pendimethalin @ 1 kg a.i./ha (88.93%) and seed treatment of *Trichoderma viride* @ 4 g/kg (75.60%). The results are quite similar with the results obtained under *in vivo* conditions.

In both the experiments *Trichoderma* gave encouraging results. The antagonistic activity of *Trichoderma* on soil borne pathogens including *R. bataticola* is well known. Kumar and Khare (1990), Haram et al (1996), Majumdar et al (1996), sindhan et al. (2002), Bandyopadhyay et al.(2003), Paul et al.(2008), Chaudhary et al.(2010) and Doley et al. (2012) are some of the supporters of this result.

Table 11. Percent Root rot at 30 DAS and 60 DAS day after sowing in Pot culture

Sr.No	Treatment	Incidence of Root Rot		Mean	% reduction over control
		30 DAS	60 DAS		
1	Control	7.33 (2.71)	7.67 (2.77)	7.5	-
2	Thiram@2g/kg +Carbendazim@1g/kg. (ST)	0.00 (0.00)	0.00 (0.00)	0.00	100
3	<i>Trichoderma</i> @4g/kg of seed(ST)	1.67 (1.28)	2.00 (1.41)	1.83	75.60
4	Pendimethalin@ 1.0kg a.i /ha(PE)	5.33 (2.31)	6.33 (2.51)	5.83	22.26
5	Thiram@2g/kg +Carbendazim@1g/kg. +Pendimethalin@1.0kg a.i /ha (PE)	0.67 (0.67)	1.00 (1.00)	0.83	88.93
6	<i>Trichoderma</i> @4g/kg of seed(ST)+Pendimethalin@ 1.0kg a.i /ha (PE)	2.33 (1.52)	2.67 (1.63)	2.5	66.67
7	<i>Trichoderma</i> @ 2.5kg/ha. (soil appli.)	1.33 (1.14)	1.67 (1.28)	1.5	80.00
8	<i>Trichoderma</i> @2.5kg/ha (Soil)+Pendimethalin@ 1.0kg a.i /ha(PE)	3.67 (1.91)	4.00 (2.00)	3.83	48.93
9	Thiram@2g/kg +Carbendazim@1g/kg. (ST)+ <i>Trichoderma</i> @2.5kg/ha (Soil)	0.33 (0.33)	0.33 (0.33)	0.33	95.00
10	Thiram@2g/kg +Carbendazim@1g/kg+ <i>Trichoderma</i> @ 2.5kg/ha +Pendimethalin@ 1.0kg a.i /ha(PE)	1.33 (1.14)	1.67 (1.28)	1.5	80.00
	'F' test	Sig	Sig	Sig	
	SE(M)±	0.16	0.13		
	CD(P=0.05)	0.50	0.39		

Values are square root transformation



A - Thiram(0.2%) 50WP+Carbendazim (0.1%) 75 WP

B - Thiram(0.2%) 50 WP+Carbendazim 0.1%)75WP+Pendimethalin 30%EC(100µl)

C – Pendimethalin 30% EC(100µl)

D – Control

CHAPTER V

SUMMARY AND CONCLUSIONS

A experiment entitled "Effect of herbicides on root rot of Chickpea caused by *Rhizoctonia bataticola*" was carried out during Rabi, 2013-2014 in the field of Pulses Research Unit and Pot culture in Department of Plant Pathology, Dr.PDKV, Akola with objective to assess the effect of herbicide on root rot of chickpea caused by *Rhizoctonia bataticola* *in vitro* and *in vivo*.

During the course of investigation observations were recorded on per cent mycelium inhibition of *Rhizoctonia bataticola* due to fungicides and herbicides at different temperature and pH, per cent mycelium inhibition of *Trichoderma* due to herbicide pendimethalin, *in vitro* and per cent root rot incidence, population of *Rhizoctonia bataticola* and population of *Trichoderma viride* *in vivo*.

The pathogenicity of *Rhizoctonia bataticola* was proved by soil inoculation method. Based on the morphological and pathogenic characteristics the isolates were identified as *Rhizoctonia bataticola*.

5.1 Effect of fungicides, herbicides and bioagents on *Rhizoctonia bataticola*.

The combination of fungicides Thiram +Carbendazim were found most effective in inhibiting cent per cent mycelial growth of *Rhizoctonia bataticola*. In combination of herbicide and fungicides Thiram+ Carbendazim +Pendimethalin also showed cent per cent inhibition of pathogen at 5.5, 6.5, 7.5 and 8.5 pH regime and all temperature ranges i.e.25°C,30°C, 35°C and 40°C.

The pre-emergence herbicide Pendimethalin alone (100µl /100 ml media) inhibited the growth of pathogen by 58.52, 55.23, 54.23 and 60.11 % at 5.5, 6.5, 7.5 and 8.5 pH at 25 ° C temps.

At 30° C the herbicide was less effective in inhibiting the growth of pathogen and recorde only 8.52, 4.85, 5.14, 10.02 %

inhibition at 5.5, 6.5, 7.5 and 8.5 pH respectively. The inhibition was 23.45, 18.60, 12.57 and 30.29 % at 5.5, 6.5, 7.5 and 8.5 pH, respectively when tested at 35°C temperature. At 40°C temperature also the herbicide became more effective and inhibited the mycelia growth of pathogen by 64.68, 65.06, 60.99, 68.75 at 5.5, 6.5, 7.5 and 8.5 pH respectively.

The herbicide pendimethalin @ 1 kg a.i./ha (100µl/100 ml) also exhibited adverse effect on *Trichoderma* and showed 77.64 per cent growth inhibition over control after 7 days of incubation period.

In vitro antagonism of *Trichoderma viride* against *R.bataticola* (dual culture technique) in herbicide added media showed inhibition of *R.bataticola* to the extent of 46.05%.whereas *Trichoderma* was also inhibited due to herbicide to the tune of 61.11%

The highest average per cent reduction in population of *R.bataticola* was observed in seed treatment of Thiram @ 2g/kg + Carbendazim @ 1 g/kg of seed + soil application of *Trichoderma* i.e. 84.75 per cent followed by soil application of *Trichoderma* alone @ 2.5 kg/ha (82.21) and application of *Trichoderma* + Pendimethalin (79.67) when observed at 15, 30 and 45 DAS.

The average population population of *Trichoderma viride* at 15,30 and 45 DAS was highest in T₇ i.e soil application of *Trichoderma* @ 2.5 kg/ha (16.5×10^4 cfu/g soil) and lowest in T₄ soil application of Pendimethalin @ 1 kg a.i./ha (1.33×10^4 cfu/g soil). All other treatments where chemicals were used showed less population of *Trichoderma* compared to control.

The treatment differences in regard to root rot incidence were significant under field conditions. Maximum root rot reduction was achieved due to combine application of Thiram@ 2 g/kg +Carbendazim @ 1 g/kg as seed treatment + soil application of *Trichoderma* @ 2.5 kg/ha (77.59%) followed by alone application of *Trichoderma* through soil (72.72%). Pendimethalin also showed to posses the antifungal activities as root rot reduction was observed to the tune of 13.31%.

In pot culture the 100 % reduction of root rot was observed in seed treatment of Thiram @ 2g/kg +Carbendazim@ 1g/kg, but when same fungicides were combined with application of *Trichoderma* through soil the reduction was slightly declined and recorded as 95 % which showed slight adverse effect of these fungicides on *Trichoderma*. This was followed by seed treatment with @ 2g/kg +Carbendazim@ 1g/kg + soil application of Pendimethalin @ 1 kg a.i./ha (88.93%) and seed treatment of *Trichoderma viride* alone @ 4 g/kg of seed (75.60%).

5.2 Conclusions

On the basis of results obtained from present investigation it can be concluded that-

- 1) Thiram @ 0.2 %, carbendazim @ 0.1% and *Trichoderma* were most effective in inhibiting mycelial growth of *Rhizoctonia bataticola*.
- 2) The pre-emergence herbicide Pendimethalin @ 100 μ l/ 100 ml media i.e 1 kg a.i./ha was most inhibitory to *Rhizoctonia bataticola* at 40°C followed by 25°C temperature at all the pH tested i.e. 5.5, 6.5, 7.5 and 8.5
- 3) Herbicide Pendimethalin @ 100 μ l/ 100 ml media i.e 1 kg a.i./ha also inhibited the growth of *Trichoderma viride* indicating that the herbicides are not compatible with *Trichoderma*.
- 4) Maximum reduction of root rot in field and pot culture was achieved due to seed treatment with Thiram @ 2 g / kg + Carbendazim @ 1 g /kg + *Trichoderma* @ 4 g /kg followed by Thiram @ 2 g / kg + Carbendazim @ 1 g /kg of seed.
- 5) Combined application of fungicides (Thiram and Carbendazim) and herbicide (Pendimethalin) may be a useful for the management of root rot disease.

CHAPTER VI

LITERATURE CITED

- Amrutha Veena., N. P. E. Reddy., B. V. B. Reddy and L. Prasanthi. 2014. Potential of *Trichoderma* spp. as biocontrol agents against *Rhizoctonia Bataticola* causing dry root rot of chickpea. Department of Plant Pathology, S. V. Agricultural College, Tirupati-517 502.
- Altman, J. and C.L. Campbell. 1977. Effect of herbicides on plant diseases. Annu. Rev. Phytopathol. 15, 361–385.
- Awadalla, O.A. and I.M. El-Refai. 1994. Effect of herbicides on the toxicity of fungicides against *Rhizoctonia solani* causing damping-off of cotton. Journal of Phytopathology. 140 (3): 187-192
- Amusa, N. A., R. U. Okechukwu and B. Akinfenwa. 2007. Reactions of cowpea to infection by *Macrophomina phaseolina* isolates from leguminous plants in Nigeria. African J. Agril. Res., 2(3): 73-75.
- Aghakhani, M., and S.C. Duubey. 2009. Determination of genetic diversity among Indian Isolates of *Rhizoctonia bataticola* causing dry root rot of chickpea. Antoine Van Leeuwenhoek, 96:607-619.
- Andrabi, Muneeb Amrish Vaid., Vijay kumar and Razdan. 2011. Evaluation of different measures to control wilt causing pathogens in chickpea. Journal of plant protection research Vol.51, No.1.
- Bandyopadhyay, S., N.D. Sharma and S. Dutta. 2003. Screening of potential *Trichoderma* strains against major root pathogens. Mannuals of plant Protection Sciences. 11:163
- Bauske, E.M. and H.W. Kirby. 1992. Effect of dinitroaniline herbicides, carboxin, pentachloronitrobenzene seed treatment, and *Rhizoctonia* disease on soybean. Plant Dis. 76, 236–239.
- Black, B.D., Russin, J.S., Griffin, J. L. and J.P. Snow. 1996. Herbicide effects on *Rhizoctonia solani* in vitro and *Rhizoctonia* foliar blight of soybean (*Glycine max*). Weed Sci. 44, 711–716.
- Bagwan, N.B. 2010. Evaluation of *Trichoderma* a compatibility with fungicides, Pesticides, organic *Trichoderma* and *Gliocladium* sp. against root knot cakes and Botanicals for integrated management of nematode (*Meloidogyne incognita*) On Sunflower, soil borne diseases of soybean (*Glycine max* (L.) Proceedings of National symposium on rational (Merrill). Int., J. Plant Prot., 3: 206-209.

- Bradley, C. A., G. L. Hartman., L. M., Wax and W. L. Pedersen. 2002. Influence of herbicides on *Rhizoctonia* root and hypocotyls rot of soybean. A Department of Crop Sciences, National Soybean Research Laboratory. *Crop Protection* 21: 679-687.
- Bowman, J.E. and J.B. Sinclair. 1989. Effect of herbicides on *Rhizoctonia* seedling disease of soybeans in glasshouse experiments. *J. Phytopathol.* 124, 267-274.
- Canaday, C.H., Helsel, D.G., and T.D. Wyllie. 1986. Effects of herbicide induced stress on root colonization of soybeans by *Macrophomina phaseolina*. *Plant Dis.* 70, 863-866.
- Choudhary, S., S. Pareek and J. Saxena. 2010. Efficacy of biocontrol agents singly and in combinations against dry root rot (*Macrophomina phaseolina*) of mungbean. *Journal of Mycology and Plant Pathology.* 40 (1):141- 144.
- Csondes, I., S. Kadlicsko and R .Gáborjányi.2007. Effect of different temperature and Culture media on the growth of *Macrophomina phaseolina*. *Commun Agric App Biol Sci.* 72(4):839-48.
- Chattannavar, S. N., S. Kulkarni., and B. M. Khadi. 2006. Effect of herbicides on *Trichoderma harzianum*. *Journal of Cotton Research and Development* .Vol. 20, (2) pp. 296-297.
- Chirame, B. B. and D. N. Padule. 2005. Effect of *Trichoderma* spp. on the growth of *Macrophomina phaseolina*, *Colletotrichum capsici* and *Fusarium moniliformae*, isolated from cotton seed. *Agric. Sci. Digest.* 25(3): 213-218.
- Chopra, B. L. and J. R. Sharma. 1986. Green house studies on evaluation of cotton varieties and fungicides against seedling mortality caused by *Rhizoctonia solani*. *Pl. Dis. Res.* 1(1-2): 83-85.
- Chauhan, M. S. 1986. Comparative efficacy of fungicides for the control of seedling Disease of cotton due to *Rhizoctonia* spp. *Indian J. Myco. and Pl. Path.* 16 (3):335-337
- Chen Hou De., Ni Gui Hua., wang Zhang Ming and Shu Zhong.2005.Effect of seven Herbicides on growth and pathogenicity of *Rhizoctonia bataticola* *cerealis* vander Hoeven, Chinese. *Journal of Yangzhou University, Agriculture and Life Sciences.* Edition.26:3, 66-69, 78.
- Desai, S.A. and S. Kulkarni. 2004. Effect of Fungicides, Insecticides and Weedicides on the Growth and Sporulation of Native *Trichoderma harzianum*. *J.Agric.Sci.* 17(1): 57-62. Department of Plant Pathology, University of Agricultural Sciences,Dharwad.

- Dissanayake N., J.W. Hoy., and J.L. Canaday. 1998. Herbicide Effects on Sugarcane Growth, *Pythium* Root Rot, and *Pythium arrhenomanes*. *Phytopathology* 88(6):530-5.
- Dhingra, O.D. and J.B. Sinclair, 1978. Variation among isolates of (*R. bataticola*) from different regions. *Phytopath.* 76:200-204.
- Doley, K. and Paramjit Kaur jite. 2012. *In-Vitro* Efficacy of *Trichoderma viride* against *Sclerotium rolfsii* and *Macrophomina phaseolina*. *Sci Biol.* 4(4):39-44.
- Dennis, C. and J. Webster. 1971. Antagonistic properties of species groups of *Trichoderma*, hyphae interaction. *Trans Br Mycol Soc* 57:363-369.
- El-Fiki, A. I. I., A. A. El-Deeb., F. G. Mohamed. and M. M. A Khalifa. 1987. Pathological and Physiological studies on some isolates of *Macrophomina phaseolina*. Department of Agric. Botany, Fac. of Agric. Moshtohor, Zagazig University (Benha Branch), Egypt. Institute of Plant Pathology, Agric. Res. Center Giza, Egypt.
- Goyal.M.K. and R.S. Mehrotra. 1981. Chemical control of dry root rot of gram caused by *R.bataticola*. *ActaBotanica India.*9:228-232.
- Gurha, S.N and R.S.Dubey. 1982. Occurrence of possible sources of resistance in Chickpea against *Sclerotium rolfsii*. *Sacc. Madras Agric.J.*70:63-64.
- Harikrishnan R. and X. B. Yang. 2002. Effects of Herbicides on Root Rot and Damping-off Caused by *Rhizoctonia solani* in Glyphosate-Tolerant Soybean. Department of Plant Pathology, Iowa State University. *Plant Dis.* 86:1369-1373.
- Hooda, I. and R.K. Grover. 1988. Effect of age, quantity of inoculum and isolates of *Macrophomina phaseolina* on the pathogens of mungbean and its control by Chemicals. *Indian Phytopathology.* 41, 107 - 117.
- Haram, S., Schikler, H. and I. Chet, 1996. Differential expression *Trichoderma harzianum* during mycoparasitism. *Phytopathology* 86 :980-985.
- Heydari, A., and I.J. Misaghi. 1998. The impact of herbicides on the incidence and development of *Rhizoctonia solani*-induced cotton seedling damping-off. *Plant Dis.* 82, 110–113.
- Haware, M.P., Y. L. Nene and M. Natarajan, 1996. The survival of *Fusarium oxysporum* f. sp. *ciceri* in soil in the absence of chickpea. *Phytopathol Mediterr* 35: 19–12.
- Jackson, C.R. 1965. Peanut kernel infection and growth in-vitro by four fungi at various temperatures. *Phytopath.*55: 46-48.

- Jha, K. M. and N.D. Sharma. 2006. Studies on factors affecting *Rhizoctonia bataticola*. *Herbicides J. of Myco. Res.* 44 (1): 73-77.
- John Priya., N. N. Tripathi and N. Kumar. 2010. Efficacy of fungicides against charcoal rot of sesame caused by *Rhizoctonia bataticola* (Taub) Butler. *Research on Crops.* 11(2): 508-510
- Jeyalakshmi, C., D. Sabitha and V. Valluvaparidasan. 1999. Studies on the seed borne mycoflora of MCU cotton cultivars, their effect and biological control. *J. Cotton Res. Dev.*, 13(1): 40-44.
- Johnson, M and K. Subramanyam, 2002. *In vitro* efficacy of fungicides against stem rot pathogen (*Sclerotium rolfsii*) of groundnut. *Manuals of Plant Protection Sciences.* 8: 255-257.
- Khan, R. A., T. A. Bhat and K. Kumar. 2012. Management of Chickpea (*Cicer arietinum* L.) Dry root Rot Caused by *Rhizoctonia bataticola* (Taub) Butler. *International Journal of Research in Pharmaceutical and Biomedical Sciences.* Department of Botany, Singhania University, Jhunjhuna, Rajasthan, India, Vol. 3 (4): 1539.
- Karthikeyan, M., K. Radhika, R. Mathiyazhagan, R. Bhaskaran, R. Samiyappan and R. Velazhahan. 2006. Induction of phenolics and defense related enzymes in coconut (*Cocos nucifera* L.) roots treated with biocontrol agents. *Braz. J. Plant Physiol.* 18:367-377.
- Kamdi, D.R., M.K. Mondhe, G. Jadesha, D. N. Kshirsagar and K. D. Thakur. 2012. Efficacy of botanicals, bio-agents and fungicides against *Fusarium Oxysporum* F. Sp. *Ciceri*, in chickpea wilt sick plot. 3 (11).
- Kaushal, R.P. 2008. Management of root rot in chickpea. *Journal of Food Legumes.* 21 (3): 178-181.
- Katan, J. and Y. Eshel, 1973. Interactions between herbicides and plant pathogen *Residue Reviews.* 45: 145-147.
- Kumar, S.M. and M.N. Khare. 1990. The studies on the antagonistic relationship of soybean spermosphere microflora with *R. bataticola* and *S. rolfsii*. *J. Boil. Control.* 4: 72-74.
- Katariya, L., V.K. Gaur and R. Sharma 2007. Assessment of genetic variability in *Rhizoctonia bataticola* infecting chickpea isolates using pathogenicity and RAPD markers. *J. Mycol. Pl. Pathol.* Vol. 37 No. 3: 491-494.
- Lambhate, S. S., G. K. Chaudhari., S. S. Mehetre and S. R. Zanjare. 2002. Biological Control of root rot of cotton caused by *Macrophomina phaseolina*, J. Maharashtra Agric. Univ., 27(1): 98

- Mohamad, said zaki Tewfik .1973. Effect of nitro herbicide on growth of soil borne fungi. Journal of weed and weed control. Pp-780-786.
- Mamta, Sharma and S.Pande.2013. Unravelling Effects of Temperature and Soil Moisture Stress Response on Development of Dry Root Rot *Rhizoctonia bataticola* (Taub.) Butler in Chickpea. American Journal of Plant Science. 4, pp. 584- 589.
- Mathur, S.C. 2006. Sclerotial root rot disease of groundnut in UP. *Current Science*. 22: 214-215.
- Mushtaq, Ahmed Reeti Singh., Shastry, P.P., Ahmed, M and R. Singh. 1997. Suppression of soil borne pathogens of soybean *in vitro* and *in vivo* by herbicides. Indian Phytopathology. 39 (3): 35-399.
- Menten, J. O. M. 1976. Efeito de alguns fungicidas. No, v. 1, n. 2, p. 57-66., 31(4): 444-448.
- Majumdar, V.L., J.R Jat and H.N Gour. 1996. Effects of biocontrol agents on the growth Of *Macrophomina phaseolina*, the incitant of blight of mothbean. Indian J. Mycol Pl. Pathol. 26:201-203
- Meyer, M. C., J. C. Silva., G. L. Maia., C. J. Bueno and N. L. Souza. 1973. Myrothecium leaf spot of cotton caused by *Myrothecium roridum*. Summa phytopathol. 32(4)
- Naseema, B., S. A. Salim., M. Priya Mohan Deepti and A .Naseema . 2005. Compatability of *Trichoderma harzianum* with botanicals and synthetic pesticides. Geobios. 32 (3): 80-82.
- Nagamma, G., M. Saifulla and T. Narendrappa. 2012. *In vitro* evaluation of fungicides, bio agents and botanicals against *Macrophomina phaseolina* (Tassi) Goid the causal agent of dry root rot in chickpea. Mysore J. Agric. Sci. 46(4): 778-783.
- Nagamani, P., Viswanath, K. and T.K. Babu. 2011. Managment of dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler in Chickpea .Biotica Vol.5 ,No.3,pp.364-369.
- Paul,Y.S., M. Devi, and A.S. Kapoor. 2008. Integrated organic management of wilt-root rot complex of pea. Journal of Mycology and Plant Pathology. 38 (3): 571-576.
- Patel, K. K. and A. J. Patel. 1990. "Meterological Correlation of Charcoal Rot of Sesamum," Indian J. of Mycology and Plant Pathology, Vol. 20, No. 1, pp. 64-65.
- Prabhu, H.V. 2009. Studies on collar rot of soybean caused *sclerotium rolfsii* sacc. M.sc. (Agri.) Thesis,university of Agricultural Sci., Dharwad,pp.74-80.

- Parab, P.B., M.P. Diwakar., U.K Sawant., and J.J. Kadam. 2009. Exploration of *Trichoderma harzianum* as antagonist against *Fusarium spp.* causing damping off and root rot disease and its sensitivity to different fungicides. Journal of Plant Disease Sciences. 4 (1): 52-56
- Peshney, N.L., R.M. Gade and K.G. Thakur. 1992. Sensitivity and adaptability of *R. bataticola* to different fungicides. J. soils & Crops. 2: 35-38.
- Patel, U.B. and A. S. Anahosur. 2000. Association of soilborne fungi infecting chickpea and correlation coefficient between frequency of fungi and other factors. Journal of mycology and plant pathology. Vol.30No.1pp.50-52.
- Rosane, F.B.T. 2013. Aveline Avozani, *In vitro* mycelial sensitivity of *Macrophomina phaseolina* to fungicides. Pesq. Agropec. Trop, Goiânia. 43: (4)460- 466.
- Russin, J.S., B.D. Black., J.L. Griffin and J.P Snow. 1995. Herbicide effects on *Rhizoctonia solani* in vitro and *Rhizoctonia* foliar blight of soybean (*Glycine max*). Weed Sci. 44, 711–716.
- Reddy, M.V., T.N. Raju and Y.L. Nene, 1990. Evaluation of chickpea lines in the all India coordinated varietal trails for wilt and dry root rot resistance. Intrn. Chickpea Newsl. 22 & 23.
- Rai, J. P., K.S. Dubey and Bijender Kumar. 2007. *In vitro* screening of different fungicides and antifungal antibiotics against *Rhizoctonia solani* causing aerial blight of soybean. Journal of Plant Disease Sciences. 2 (1): 54-56.
- Rangaswami, G. 1958. Diseases of crop Plants in India..24:85 prentice hall of India,Pvt.Ltd.,New Delhi.
- Saxena, D., A.K. Tewari and D. Rai. 2014. *In vitro* Effect of Some Commonly Used Fungicides, Insecticides and Herbicides for Their Compatibility with *Trichoderma harzianum* PBT23. World Applied Sciences Journal. Department of Plant Pathology.
- Schmitz, H. 1930. A suggested toximetric method for food preservation. Indust. and Engin. Chem. Analyst. Ed. 4: 361-365.
- Singh, S.K. and Nene, Y.L. 1990. Influence of cropping system on *R. bataticola* population in soil. Plant Dis. 74: 812-814.
- Sindhan, G.S., I. Hooda and S.S. Karwasara 2002. Biological control of dry root rot of chickpea caused by *Rhizoctonia bataticola*. Pl.Dis.Res.17 (1):68-71.
- Sundravadana, S. 2002. Management of blackgram (*Vigna mungo* (L.) Hepper) root rot (*Macrophomina phaseolina* (Tassi.) Goid with bioagents and nutrient. M.Sc (Ag) Thesis, Tamil Nadu Agricultural University, Coimbatore, India.

- Singh, S.B. and G. Kulshrestha. 1991. Microbial degradation of pendimethalin, Division of Agricultural Chemicals, Indian Agricultural Research Institute, New Delhi. 26 (3): 309-321.
- Saxena, H.K. and O. Vaartaja. 1961. Taxonomy, morphology, and pathogenicity of *Rhizoctonia* species from forest nurseries. Canadian J. of Botany.39(3):627-647, 10.1139/b61-051.
- Suryawanshi, A. P., D. D. Gore., D. B. Gawade., A. K. Pawar and A. G. Wadje. 2008. Efficiency of fungicides against *Macrophomina* Blight of mungbean. J .Pl. Dis .Sci. 3(1): 40-42
- Srivastav, A. K. Arora, D. K., Gupta, S., Pandey, R. R. and Lee, M. 1976. Diversity of potential microbial parasites colonizing sclerotia of *Macrophomina phaseolina* in soil. Biol. Fertil. Soils 22: 136–140.
- Smith, R.S. 1964. Effect of diurnal temperature fluctuation on linear growth rate of *R. bataticola* in culture. Phytopath.54: 849-852.
- Taya, R.S., N.N.Tripathi and M.S. Panwar. 1990. Influence of texture and nutritional status of soil on the efficacy of fungicides for the control of dry root rot of chickpea. Indian J. Mycol. Pl.Pathol.20:14-20.
- Tiwari, R.K. S. and A. Singh. 2004. Efficacy of fungicides on *Rhizoctonia solani* and *Sclerotium rolfsii* and their effect on *Trichoderma harzianum* and *Rhizobium leguminosarum*. J. of Mycology and Plant Pathology. 30 (2): 134-139.
- Uppal, B.N., K.G.Kolhatkar and M.K.Patel.1936.Dry root rot and wilt disease of chickpea. India J.Agric.Sci.6:pp.1323-1334.
- Vijay Mohan., Prasad S. M., Barnwal, M.K. and Kudada, N. 2006. Fungicidal management of dry root rot disease and yield of chickpea. J. of Applied Biology. 16 (1/2): 42-44.
- Vanita Salunkhe, Sarika Armarkar and R.W. Ingle. 2009. Effect of nutritional and physiological factors on growth and sclerotial formation of *R.bataticola* butler isolates.J.Pl.Dis.Vol.4 (1)2009:44-48.
- Xue, G. and Allen .2002. Evolution of *Tricoderma* sp. *Rhizoctonia bataticola* phytopathology. vol. 93: 325-325.
- Zidan, Z. H., S.M. Dahroug, N.A .El-Safty., A.F.Shehata. and S.S. Ramses. 1998. Factors influencing the fungicidal activities of certain fungicides, herbicides and their binary mixtures against soil-borne.

VITA

1. Name of student : **Borkar Sima Hivraj**
2. Date of Birth : 21st march 1990
3. Name of the College : Post Graduate Institute,
Dr. Panjabrao Deshmukh
Krishi Vidyapeeth, Akola
4. Residential Address : At- Post –Sendurwafa , Tal.sakoli,
Dist. –Bhandara.
Email simaborkar77@gmail.com,
Mob. No-8484985736
5. Academic Qualification :

Sr. No.	Name of Degrees awarded	Year in which obtained	Division/ Class	Name of awarding University	Subjects
1)	B.Sc. (Agri.)	2012	First	Dr. PDKV Akola	Agriculture and allied subject

6. Research papers published (if any) : NIL
7. Field of Interest : Research and Development
(in which you desire to work)

Place : Akola

Date : / /2014


Signature of Student

(Borkar Sima Hivraj)

APPENDIX-I

Media used and their composition

Potato Dextrose agar medium

Composition	Gram/litre
Peeled potato	200
Dextrose	20.00
Agar-Agar	20.00
Distilled Water	1000 ml

APPENDIX-II

Fungicides used with their trade names

Trade name	Common name	Manufacturer(s)
Bavistin	Carbenadazim	BASF india limited
TMTD	Thiram	BASF india limited

APPENDIX-III

Herbicides used with their trade names

Trade name	Common name
Stomp	Pendimethalin 30%EC

APPENDIX-IV

Bioagent

Trichoderma viride



